

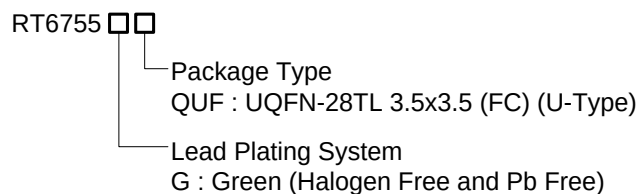
TFT LCD Integrated Power Module for Gaming Notebook

General Description

The RT6755 is an I2C interface programmable power management IC. The IC includes one synchronous boost converters for PAVDD, one synchronous NAVDD buck-boost, one SIBO (Single Inductor Bipolar Output) converter for VGH and VGL, one high performance VCOM with 8-bit Calibrator, two GMA1/2 and one RESET voltage detector with available in a UQFN-28TL 3.5x3.5 (FC) package, this device is suitable for gaming notebook TFT-LCD panel.

The IC can operate from 2.5V to 5V input voltages. High switching frequency operation allows a compact PCB circuit area with small inductors and capacitors. Current-limit functions are provided for all internal-switch converters, and output-fault shutdown protects all converters against output-fault conditions. Programmable soft-start functions for all output voltage to limit input inrush current during startup.

Ordering Information

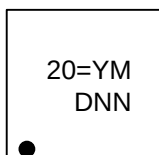


Note :

Richtek products are :

- ▶ RoHS compliant and compatible with the current requirements of IPC/JEDEC J-STD-020.
- ▶ Suitable for use in SnPb or Pb-free soldering processes.

Marking Information



20= : Product Code
YMDNN : Date Code

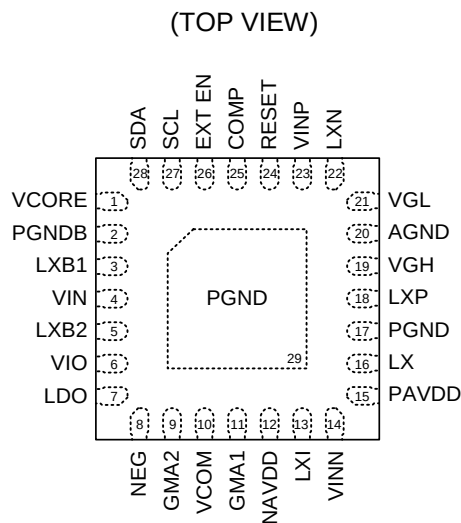
Features

- 2.5V to 5V Input Supply Range
- I²C Interface
- PAVDD Synchronous Boost Converter
 - ▶ Programmable Output Range from 4V to 7V
 - ▶ Output Current Capability up to 400mA
 - ▶ Current Limitation Protection
- NAVDD Synchronous Inverting Converter
 - ▶ Programmable Output Voltage -4V to -7V
 - ▶ Output Current Capability up to 400mA
 - ▶ Current Limitation Protection
- VGH / VGL Synchronous SIBO Converter
 - ▶ VGH Programmable Output Range from 6V to 30V
 - ▶ VGL Programmable Output Range -6V to -18V
 - ▶ VGH and VGL Output Current Capability up to 20mA
- VCORE and VIO Synchronous Buck Converter
 - ▶ VCORE Programmable Output Range from 0.8V to 2V
 - ▶ VIO Programmable Output Range from 1V to 2.5V
- Low Dropout Voltage Linear Regulator
- VCOM 8bits Programmable Output Voltage
- GMA1/2 Programmable Gamma Output Voltage
- Programmable Voltage Detector
- Built in UVLO, UVP, OVP, SCP and OTP protection
- RoHS Compliant and Halogen Free

Applications

- TFT-LCD Notebook
- TFT-LCD Tablet

Pin Configuration



UQFN-28L 3.5x3.5 (FC)

Typical Application Circuit

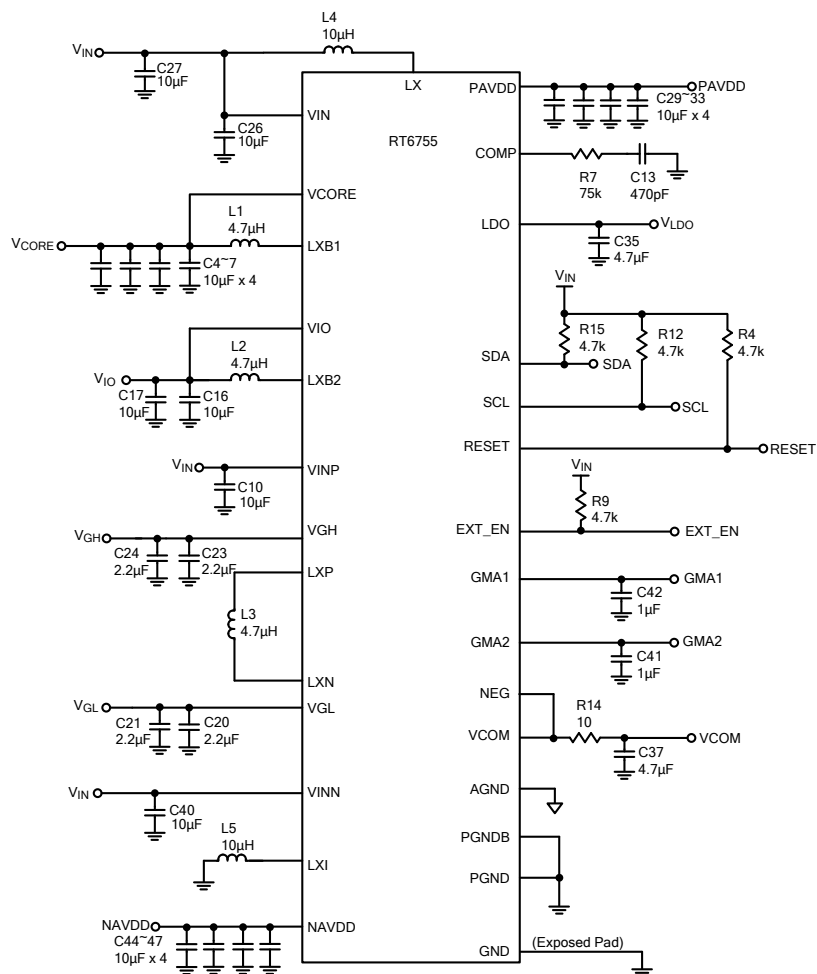


Figure 1. General Application Circuit

Table 1. Component List of Evaluation Board

Components	Part Number	Description	Package	Supplier
C35, C37	C1608X5R0J475K080AB	4.7uF/6.3V/X5R/0603	1.6mmx0.8mmx0.8mm	TDK
C4~C7,C10,C16, C17,C26,C27, C40	GRM188R60J106ME47	10uF/6.3V/X5R/0603	1.6mmx0.8mmx0.8mm	MURATA
C41, C42	GRM155R61A105ME15J	1uF/10V/X5R/0402	1.0mmx0.5mmx0.5mm	MURATA
C44~C47, C29~C33	GRM188R61C106KAAL	10uF/16V/X5R/0603	1.6mmx0.8mmx0.8mm	MURATA
C20, C21	GRM155R61E225KE11	2.2uF/25V/X5R/0402	1.0mmx0.5mmx0.5mm	MURATA
C23, C24	GRM188R61H225KE11	2.2uF/50V/X5R/0603	1.6mmx0.8mmx0.8mm	MURATA
L4, L5	HBMQ061T100MST	10uH	6.4mmx6.4mmx1.0mm	CYNTEC
L1, L2, L3	HTTH25201T4R7MSR	4.7uH	2.5mmx2.0mmx1.0mm	CYNTEC

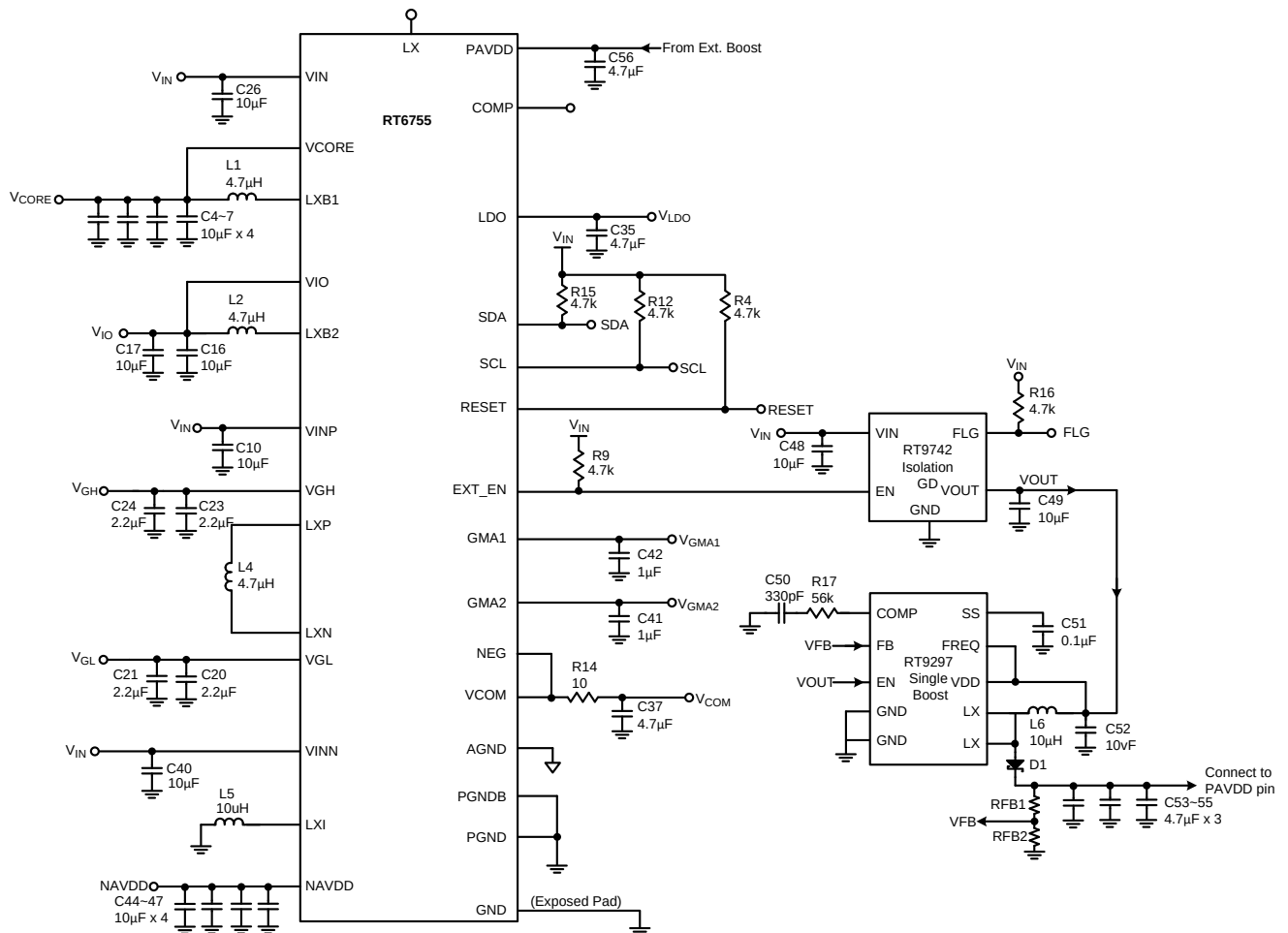


Figure 2. The Application Circuit of Using External PAVDD

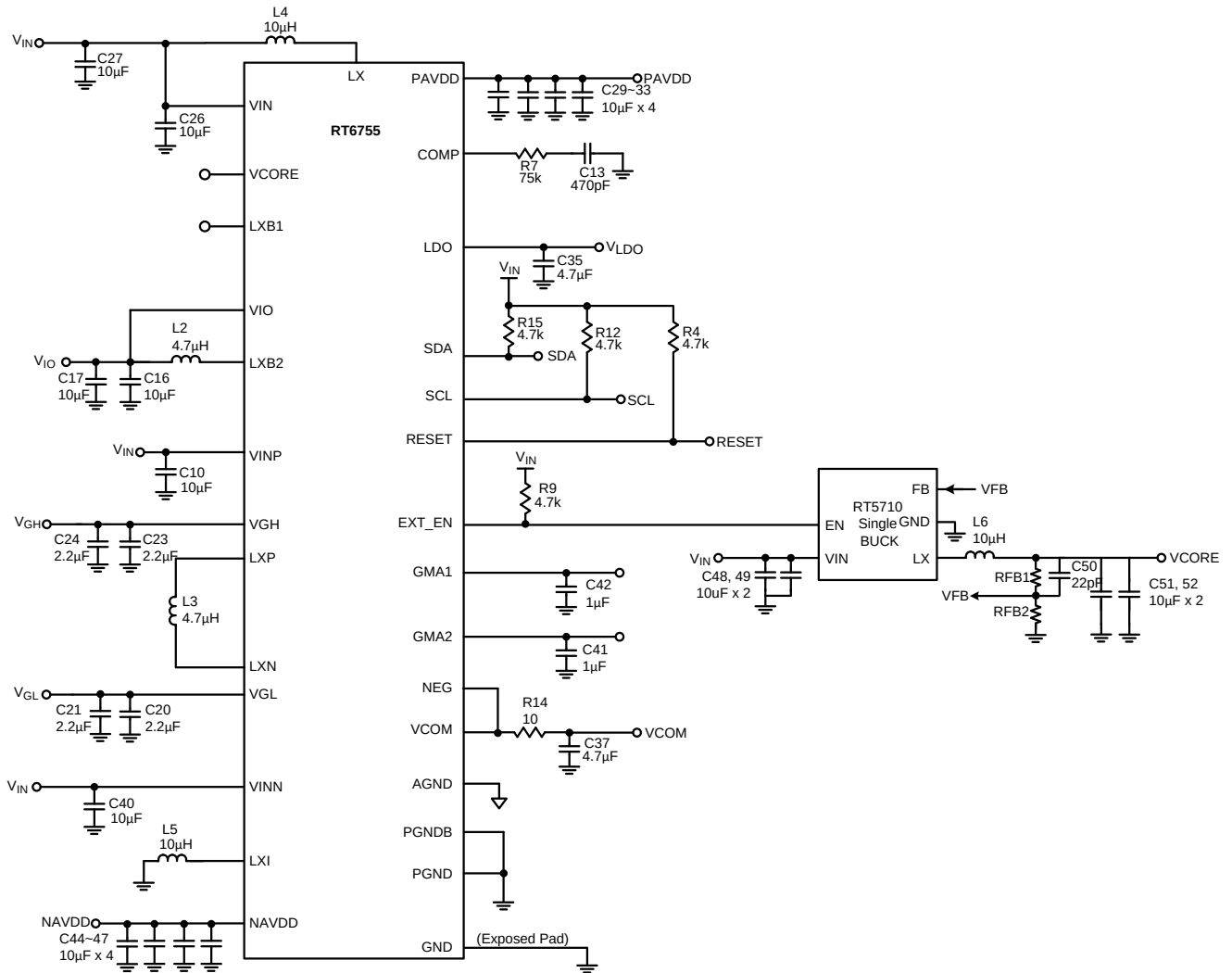


Figure 4. The Application Circuit of Using External Vcore

Timing Diagram

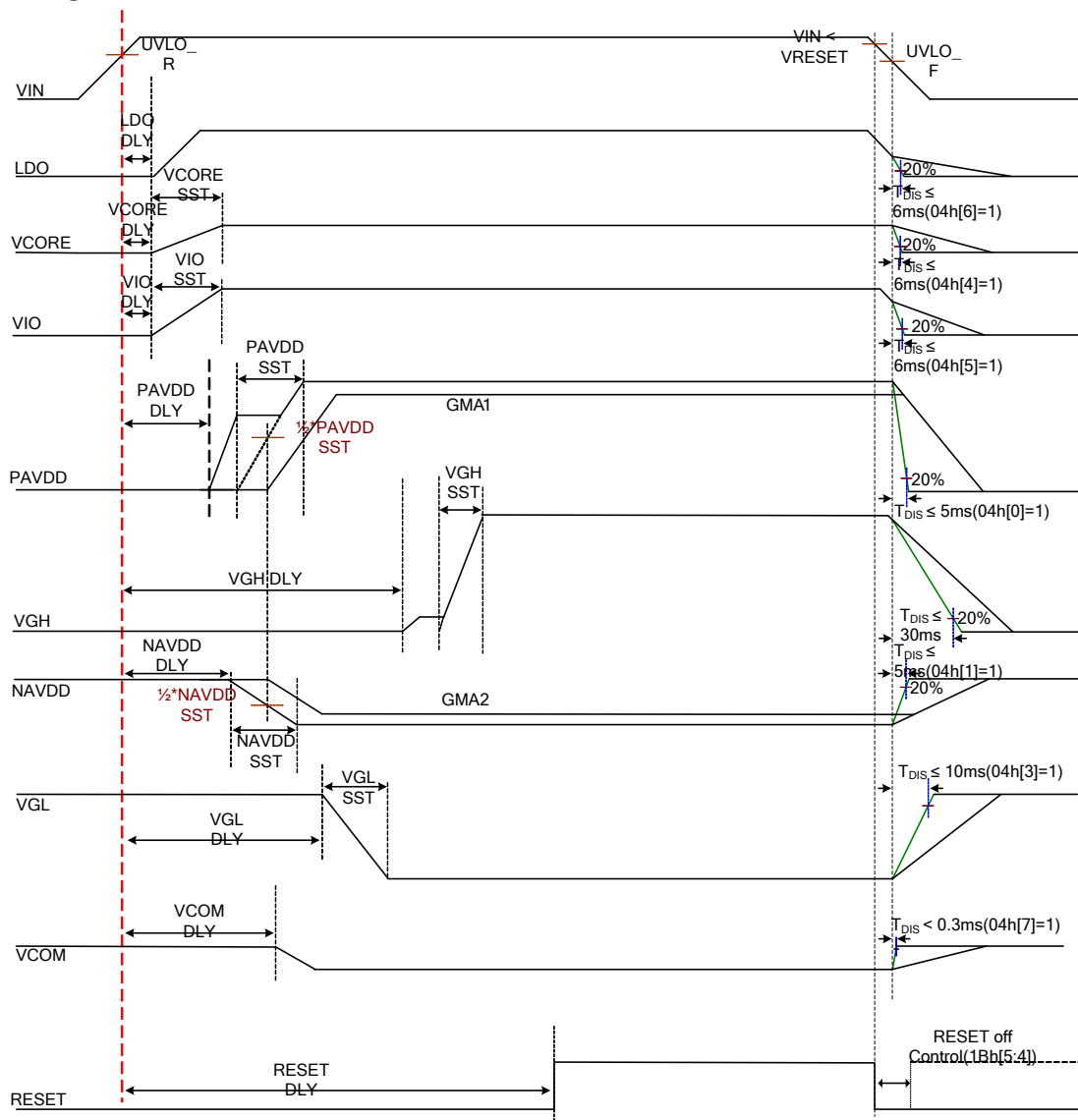


Figure 5. Timing Diagram

Note 1. $VGH / VGL DLY \geq (PAVDD DLY + PAVDD SST)$ when SIBO power source come from PAVDD.

Note 2. $VCOM DLY \geq (PAVDD DLY + PAVDD SST) \& (NAVDD DLY + NAVDD SST)$.

Note 3. $(VGL DLY + SST) \leq VGH DLY$ or $(VGH DLY + SST) \leq VGL DLY$, the soft-start of VGH and VGL can not occur at the same time due to the limitation of SIBO topology.

Note 4. When PAVDD use external converter to supplied, the PAVDD SST setting $\geq$ the external PAVDD soft-start time to make GMA1 normal operation.

Note 5. When NAVDD use external converter to supplied, the NAVDD SST setting $\geq$ the external NAVDD soft-start time to make GMA2 normal operation.

Note 6. The power-on sequence setting and channel on/off can't be changed during the power-on sequence.

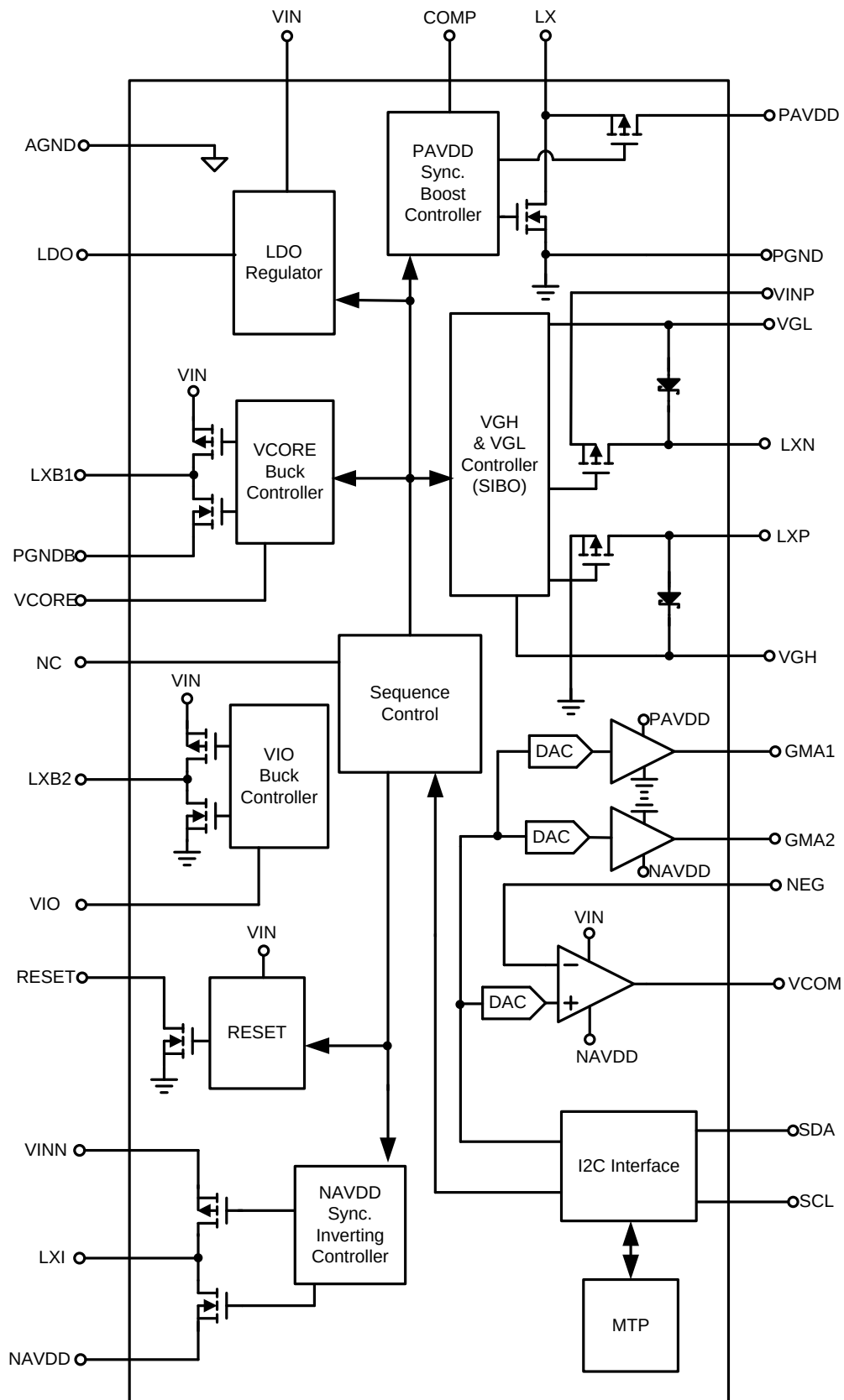
Note 7. Before the channels soft-start finish, the output should not be added load for preventing the power-up unsuccessfully.

Note 8. The resistance between power supply and VIN pin of the RT6755 chip should be smaller than 0.5 ohm to prevent the power-on un-successfully.

Functional Pin Description

Pin No.	Pin Name	Pin Function
1	VCORE	VCORE buck converter output.
2	PGNDB	Power ground
3	LXB1	Switching node of VCORE buck converter.
4	VIN	Power supply input of IC
5	LXB2	Switching node of VIO buck converter.
6	VIO	VIO buck converter output.
7	LDO	LDO output.
8	NEG	Inverting input of VCOM-OP
9	GMA2	Negative GMA2 OP output
10	VCOM	VCOM Op-Amp output.
11	GMA1	Positive GMA1 OP output
12	NAVDD	NAVDD buck-boost converter output.
13	LXI	Switching node of NAVDD buck-boost converter.
14	VINN	Power supply input of NAVDD buck-boost converter
15	PAVDD	PAVDD boost converter output.
16	LX	Switching node of PAVDD boost converter.
17	PGND	Power ground of PAVDD boost converter.
18	LXP	Switching node of VGH SIBO converter.
19	VGH	VGH SIBO converter output.
20	AGND	Analog ground.
21	VGL	VGL SIBO converter output.
22	LXN	Switching node of VGL SIBO converter.
23	VINP	Power supply input of SIBO converter
24	RESET	Output of VIN voltage detection function
25	COMP	Compensation pin of PAVDD boost converter
26	EXT EN	An external enable signal of external PAVDD, NAVDD or VCORE.
27	SCL	Serial clock line of I ² C
28	SDA	Serial data line of I ² C

Functional Block Diagram



Absolute Maximum Ratings (Note 8)

- VIN, VINP, VINN, PAVDD, LX, GMA1 to GND-----0.3V to 10V
- RESET, SDA, SCL, EXT_EN, COMP to GND -----0.3V to 7V
- LDO, VCORE, VIO, LXB1, LXB2 to GND -----0.3V to 7V
- LXP, VGH to GND -----0.3V to 32V
- LXN to GND -----(VGL + 0.3) to (VINP – 0.3V)
- VGL to GND -----20V to 0.3V
- GMA2, NAVDD to GND -----10V to 0.3V
- LXI to GND----- (NAVDD + 0.3V) to (VINN – 0.3V)
- NEG to GND-----8V to 8V
- VCOM to GND ----- (NAVDD + 0.3V) to (VIN – 0.3V)
- PGND, AGND to PGND -----0.3V to 0.3V
- Power Dissipation, P_D @ $T_A = 25^\circ\text{C}$
 UQFN-28TL 3.5x3.5 (FC) -----3.2W
- Package Thermal Resistance (Note 9)
 UQFN-28TL 3.5x3.5 (FC), θ_{JA} -----31.2°C/W
 UQFN-28TL 3.5x3.5 (FC), θ_{JC} -----2.2°C/W
- Storage Temperature Range -----65°C to 150°C
- Lead Temperature (Soldering, 10 sec.)-----260°C
- ESD Susceptibility (Note 10)
 HBM (Human Body Mode) -----2kV

Recommended Operating Conditions (Note 11)

- Supply Input Voltage, VIN----- 2.5V to 5V
- Ambient Temperature Range -----40°C to 85°C
- Junction Temperature Range -----40°C to 125°C

Electrical Characteristics

(VIN = 3.3V, PAVDD = 6V, NAVDD = -6V, VGH = 18V, VGL = -14V, VCore = 1.1V, VIO = 1.8V, VCOM = -1V, TA = 25°C, unless otherwise specified)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
General						
VIN Supply Range	VIN		2.5	--	5	V
VIN OVP Threshold	VIN_OVP		--	6.8	--	V
VIN Under-voltage-lockout Falling	VIN_UVLO_F	VIN falling, turn-off IC	--	1.87	--	V
VIN Under-voltage-lockout Hysteresis	VIN_UVLO_HYS	Hys = VIN_UVLO_R-VIN_UVLO_F	--	0.4	--	V
VIN Quiescent Current	IVIN_NSW	LX not switching for all channel	--	1.6	--	mA
	IVIN_SW	LX switching for all channel	--	2.4	--	mA
Thermal Shutdown	TSD	Temperature rising	--	160	--	°C
	TSD_HYS	Hysteresis	--	25	--	°C
I²C Protocol						
High-Level Input Voltage	I ² C_VIH	SCL, SDA	1.0	--	--	V
Low-Level Input Voltage	I ² C_VIL	SCL, SDA	--	--	0.4	V
SCL Clock Frequency	I ² C_FCLK		10	400	1000	kHz
Repeat Start Condition Setup Time	I ² C_TSU_STA		260	--	600	ns
Start Condition Hold Time	I ² C_THD_STA		260	--	600	ns
Stop Condition Setup Time	I ² C_TSU_STO		260	--	600	ns
Bus Free Time Between Stop to Start	I ² C_TBUF		500	--	600	ns
Low Period of the SCL Clock	I ² C_TLOW		500	--	600	ns
High Period of the SCL Clock	I ² C_THIGH		260	--	600	ns
Data Set-up Time	I ² C_TSU_DAT		50	--	570	ns
Data Hold Time	I ² C_THD_DAT		0	--	570	ns
Timeout	I ² C_TOUT		--	50	--	ms
PAVDD Boost Converter						
Output Voltage Range	VPAVDD	VPAVDD = 05h[5:0] = 0x00 to 0x3C	4.0	--	7.0	V
Min. Output Voltage	VPAVDD_MIN	VPAVDD = 05h[5:0] = 0x00 to 0x0F	VIN + 0.3	--	--	V
		VPAVDD = 05h[5:0] = 0x10 to 0x3C	VIN + 1.5	--	--	
Default Output Voltage	VPAVDD_6V	VPAVDD = 05h[5:0] = 0x28 = 6V	5.94	6	6.06	V
Output Accuracy	VPAVDD_ACC	VPAVDD = 4V to 7V	-1	--	1	%

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Switch Frequency Range	PAVDD_FSW	PAVDD_FSW = 10h[2:0] = 0x00 to 0x05 = 600k / 715k / 800k / 900k / 1000k / 1200k	600	--	1200	kHz
Default Setting Switching Frequency	PAVDD_FSW_1 200k	PAVDD_FSW = 10h[2:0] = 0x05 = 1200k	1080	1200	1320	kHz
Switch Frequency Accuracy	PAVDD_FSW_A CC_715k	PAVDD_FSW = 10h[2:0] = 0x01 = 715k	-5	--	5	%
	PAVDD_FSW_A CC	PAVDD_FSW = 600k / 800k / 900k / 1000k / 1200k	-10	--	10	%
Maximum Duty	PAVDD_DMAX		--	90	--	%
Minimum On Time	PAVDD_MOT_4 .8V	PAVDD ≥ 4.8V	--	220	--	ns
	PAVDD_MOT_4 .75V	PAVDD < 4.8V	--	70	--	ns
Maximum Output Current Capability	IPAVDD_MAX	VIN = 3V, PAVDD = 7V	400	--	--	mA
Line Regulation	PAVDD_LIR	VIN = 2.5V to 5V	-0.3	--	0.3	%/V
Load Regulation	PAVDD_LOR	VIN = 3.3V, IPAVDD = 10mA to 400mA	--	75	--	mV/A
High-Side MOSFET On-Resistance	PAVDD_RON_HS	IPAVDD = 100mA	--	150	--	mΩ
Low-Side MOSFET On-Resistance	PAVDD_RON_LS	IPAVDD = 100mA	--	85	--	mΩ
LX Leakage Current	PAVDD_LX_ILK	VLX = 6V	--	--	1	μA
Current Limit Peak Range	PAVDD_ILIM_P EAK	PAVDD_ILIM_PEAK = 10h[6:5] = 0x00 to 0x03 = 1A / 1.5A / 2A / 2.5A	1	--	2.5	A
Default Current Limit Peak Accuracy	PAVDD_ILIM_P EAK_ACC	VIN = 2.8V to 5V.	--	1.5	--	A
Negative Current Limit Peak	PAVDD_ILIM_N EG_PEAK		--	1	--	A
Under Voltage Protection Threshold	PAVDD_UVP		--	80	--	%
Under Voltage Protection Delay Time	PAVDD_UVP_D LY		--	60	--	ms
Short Circuit Protection Threshold	PAVDD_SCP		--	1.25	--	V
Short Circuit Protection Hysteresis	PAVDD_SCP_H YS		--	92	--	mV
Over Voltage Protection Threshold	PAVDD_OVP	PAVDD rising	--	7.5	--	V
Power on/off Discharge Ron	PAVDD_RDIS	Before PAVDD soft-start and After UVLO should be turn on.	--	42	--	Ω
Power off Discharge Time	PAVDD_DIST	Discharge function enable, COUT = 40uF, Time = 100% to 20%	--	5	--	ms
Power-on Soft-start Time Range	PAVDD_SST	PAVDD_SST = 11h[6:4] = 0x00 to 0x07	2	--	16	ms
Default Power-on Soft-start Time	PAVDD_SST_4 MS	PAVDD_SST = 11h[6:4] = 0x01	3.6	4	4.4	ms
Power on Delay Time Range	PAVDD_DLY	PAVDD_DLY = 11h[3:0] = 0x00 to 0x0F	0	--	45	ms

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Default Power on Delay Time	PAVDD_DLY_1 0MS	PAVDD_DLY = 11h[3:0] = 0x02	9	10	11	ms
Pre-charge Current	PAVDD_IPC	C _{OUT} = 40uF	--	350	--	mA
VCORE Buck Converter						
Output Voltage Range	VCORE	VCORE = 09h[5:0] = 0x00~0x3C	0.8 0.2*VIN	--	2.0	V
Default Output Voltage	VCORE_1.1V	VCORE = 09h[5:0] = 0x0F	1.089	1.1	1.111	V
Output Accuracy	VCORE_ACC	VCORE = 0.8V to 2V	-1	--	1	%
Switch Frequency Range	VCORE_FSW	VCORE_FSW = 17h[2:0] = 0x00 to 0x05 = 600k / 715k / 800k / 900k / 1000k / 1200k	600	--	1200	kHz
Default Setting Switching Frequency	VCORE_FSW_1 200k	VCORE_FSW = 17h[2:0] = 0x05 = 1200k	1080	1200	1320	kHz
Switch Frequency Accuracy	VCORE_FSW_A CC_715k	VCORE_FSW = 17h[2:0] = 0x01 = 715k	-5	--	5	%
	VCORE_FSW_A CC	VCORE_FSW = 600k / 800k / 900k / 1000k / 1200k	-10	--	10	%
Maximum Duty	VCORE_DMAX		--	100	--	%
Minimum On Time	VCORE_MOT	VCORE ≥ VIN - 4.2V	--	130	--	ns
Maximum Output Current Capability	ICORE_MAX	VIN = 3.3V, VCORE = 1.2V	700	--	--	mA
Line Regulation	VCORE_LIR	VIN = 2.5V to 5V	-0.3	--	0.3	%/V
Load Regulation	VCORE_LOR	VIN = 3.3V, I _{CORE} = 10 to 800mA	--	50	--	mV/A
High-Side MOSFET On-Resistance	VCORE_RON_ HS	I _{CORE} = 100mA	--	135	--	mΩ
Low-Side MOSFET On-Resistance	VCORE_RON_ LS	I _{CORE} = 100mA	--	100	--	mΩ
LXB1 Leakage Current	VCORE_LXB1_ LK	V _{LXB1} = 0V	--	--	1	μA
Current Limit Peak	VCORE_ILIM_P EAK_ACC	VIN = 2.8V to 5V	--	1.5	--	A
Negative Current Limit Peak	VCORE_ILIM_N EG		350	--	--	mA
Under Voltage Protection Threshold	VCORE_UVP		--	80	--	%
Under Voltage Protection Delay Time	VCORE_UVP_ DLY		--	60	--	ms
Short Circuit Protection Threshold	VCORE_SCP		--	20	--	%
Power off Discharge Ron	VCORE_RDIS	Before VCORE soft-start and After UVLO_F should be turn on.	--	50	--	Ω
Power off Discharge Time	VCORE_DIST	Discharge function enable, C _{OUT} = 40uF, Time = 100% to 20%	--	6	--	ms
Power-on Soft-start Time Range	VCORE_SST	VCORE_SST = 18h[5:4] = 0x00 to 0x03	2	--	8	ms
Default Power-on Soft-start Time	VCORE_SST_4 MS	VCORE_SST = 18h[5:4] = 0x01	3.6	4	4.4	ms
Power-on Delay Time Range	VCORE_DLY	VCORE_DLY = 18h[3:0] = 0x00 to 0x0F	0	--	45	ms

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Default Power-on Delay Time	VCORE_DLY_5 MS	VCORE_DLY = 18h[3:0] = 0x01	4.5	5	5.5	ms
VIO Buck Converter						
Output Voltage Range	VIO	VIO = 0Ah[4:0] = 0x00~0x1E	1.0	--	2.5	V
Default Output Voltage	VIO_1.8V	VIO = 09h[5:0] = 0x10	1.782	1.8	1.818	V
Output Accuracy	VIO_ACC	VIO = 1V to 2.5V	-1	--	1	%
Switch Frequency Range	VIO_FSW	VIO_FSW = 19h[2:0] = 0x00 to 0x05 = 600k / 715k / 800k / 900k / 1000k / 1200k	600	--	1200	kHz
Default Setting Switching Frequency	VIO_FSW_1200k	VIO_FSW = 19h[2:0] = 0x05 = 1200k	1080	1200	1320	kHz
Switch Frequency Accuracy	VIO_FSW_ACC_715k	VIO_FSW = 19h[2:0] = 0x01 = 715k	-5	--	5	%
	VIO_FSW_ACC	VIO_FSW = 600k / 800k / 900k / 1000k / 1200k	-10	--	10	%
Maximum Duty	VIO_DMAX		--	100	--	%
Minimum On Time	VIO_MOT		--	160	--	ns
Maximum Output Current Capability	IIO_MAX	VIN = 3.3V, VIO = 1.8V	400	--	--	mA
Line Regulation	VIO_LIR	VIN = 2.5V to 5V	-0.3	--	0.3	%/V
Load Regulation	VIO_LOR	VIN = 3.3V, IIO = 10~400mA	--	50	--	mV/A
High-Side MOSFET On-Resistance	VIO_RON_HS	IIO = 100mA	--	175	--	mΩ
Low-Side MOSFET On-Resistance	VIO_RON_LS	IIO = 100mA	--	200	--	mΩ
LXB2 Leakage Current	VIO_LXB2_ILK	VLXB2 = 0V	--	--	1	μA
Current Limit Peak	VIO_ILIM_PEAK_ACC	VIN = 2.8V to 5V	--	1.2	--	A
Negative Current Limit Peak	VIO_ILIM_NEG		360	--	--	mA
Under Voltage Protection Threshold	VIO_UVP		--	80	--	%
Under Voltage Protection Delay Time	VIO_UVP_DLY		--	60	--	ms
Short Circuit Protection Threshold	VIO_SCP		--	20	--	%
Power off Discharge Ron	VIO_RDIS	Before VIO soft-start and After UVLO_F should be turn on.	--	100	--	Ω
Power off Discharge Time	VIO_DIST	Discharge function enable, COUT = 40uF, Time = 100% to 20%	--	6	--	ms
Power-on Soft-start Time Range	VIO_SST	VIO_SST = 1Ah[5:4] = 0x00 to 0x03	2	--	8	ms
Default Power-on Soft-start Time	VIO_SST_4MS	VIO_SST = 1Ah[5:4] = 0x01	3.6	4	4.4	ms
Power-on Delay Time Range	VIO_DLY	VIO_DLY = 1Ah[3:0] = 0x00 to 0x0F	0	--	45	ms

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Default Power-on Delay Time	VIO_DLY_10MS	VIO_DLY = 1Ah[3:0] = 0x02	9	10	11	ms
LDO Regulator						
Output Voltage Range	VLDO	VLDO = 0Bh[3:0] = 0x00~0x0C	1.8	--	3	V
Default Output Voltage	VLDO_2.5V	VLDO = 0Bh[3:0] = 0x07	2.475	2.5	2.525	V
Output Accuracy	VLDO_ACC	T _A = 25°C, VLDO = 1.8V to 3V	-1	--	1	%
Dropout Voltage	LDO_DROP	VIN = 2.8V, VLDO = 2.8V, ILDO = 100mA	--	0.2	--	V
Current Limit	LDO_ILIM	VIN = 3.3V, VLDO = 2.5V	--	0.28	--	A
Maximum Output Current Capability	ILDO_MAX	VIN = 3.3V, VLDO = 2.5V	100	--	--	mA
Line Regulation	LDO_LIR	VIN = 2.5V~5V	-1	--	1	%
Load Regulation	LDO_LOR	VIN = 3.3V, VLDO = 2.5V, 1mA < ILDO < 100mA	-1	--	1	%
Under Voltage Protection Threshold	LDO_UVP		--	60	--	%
Under Voltage Protection Delay Time	LDO_UVP_DLY		--	60	--	ms
Power off Discharge Ron	LDO_RDIS	Before LDO soft-start and After UVLO_F should be turn on.	--	200	--	Ω
Power off Discharge Time	LDO_DIST	Discharge Function enable, C _{OUT} = 4.7uF, Time = 100%~20%	--	6	--	ms
Power-on Delay Time Range	LDO_DLY	LDO_DLY = 0Bh[7:4] = 0x00 to 0x0F	0	--	45	ms
Default Power-on Delay Time	LDO_DLY_10MS	LDO_DLY = 0Bh[7:4] = 0x02	9	10	11	ms
NAVDD Inverting Converter						
Output Voltage Range	VNAVDD	VNAVDD = 06h[5:0] = 0x00~0x1E	-4	--	-7	V
Default Output Voltage	VNAVDD_-6V	VNAVDD = 06h[5:0] = 0x14 = -6V	-6.06	-6	-5.94	V
Output Accuracy	VNAVDD_ACC	VNAVDD = -4V to -7V	-1	--	1	%
Switch Frequency Range	NAVDD_FSW	NAVDD_FSW = 13h[2:0] = 0x00 to 0x05 = 600k / 715k / 800k / 900k / 1000k / 1200k	600	--	1200	kHz
Default Setting Switching Frequency	NAVDD_FSW_1200k	NAVDD_FSW = 13h[2:0] = 0x05 = 1200k	1080	1200	1320	kHz
Switch Frequency Accuracy	NAVDD_FSW_A CC_715k	NAVDD_FSW = 13h[2:0] = 0x01 = 715k	-5	--	5	%
	NAVDD_FSW_A CC	NAVDD_FSW = 600k / 800k / 900k / 1000k / 1200k	-10	--	10	%
Maximum Duty	NAVDD_DMAX		--	90	--	%
Minimum On Time	NAVDD_MOT		--	350	--	ns
Maximum Output Current Capability	INAVDD_MAX	VIN = 3V, NAVDD = 7V	400	--	--	mA
Line Regulation	NAVDD_LIR	VIN = 2.5V to 5V	-0.3	--	0.3	%/V

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Load Regulation	NAVDD_LOR	VIN = 3.3V, I _{NAVDD} = 10mA to 400mA	--	80	--	mV/A
High-Side MOSFET On-Resistance_Big	NAVDD_RON_H S_B	I _{NAVDD} = 100mA	--	330	--	mΩ
Low-Side MOSFET On-Resistance_Big	NAVDD_RON_L S_B	I _{NAVDD} = 100mA	--	210	--	mΩ
High-Side MOSFET On-Resistance_Small	NAVDD_RON_H S_S	I _{NAVDD} = 100mA	--	150	--	mΩ
Low-Side MOSFET On-Resistance_Small	NAVDD_RON_L S_S	I _{NAVDD} = 100mA	--	60	--	mΩ
LXI Leakage Current	NAVDD_LXI_IL K	NAVDD = -6V, V _{LXIN} = 3.3V	--	--	1	μA
Current Limit Peak Range	NAVDD_ILIM_P EAK	NAVDD_ILIM_PEAK = 13h[6:5] = 0x00 to 0x03 = 1.5A / 2A / 2.5A / 3A	1.5	--	3	A
Default Current Limit Peak Accuracy	NAVDD_ILIM_P EAK_ACC	VIN = 2.8V to 5V, NAVDD_ILIM_PEAK = 13h[6:5] = 0x02 = 2.5A	--	1.5	--	A
Negative Current Limit Peak	NAVDD_ILIM_N EG_PEAK		400	--	--	mA
Under Voltage Protection Threshold	NAVDD_UVP		--	80	--	%
Under Voltage Protection Delay Time	NAVDD_UVP_D LY	Trigger base on UVP	--	60	--	ms
Short Circuit Protection Threshold	NAVDD_SCP		--	20	--	%
Power on/off Discharge Ron	NAVDD_RDIS	Before NAVDD soft-start and After UVLO_F should be turn on.	--	42	--	Ω
Power off Discharge Time	NAVDD_DIST	Discharge function enable, C _{OUT} = 40uF, time = 100% to 20%	--	5	--	ms
Power-on Soft-start Time Range	NAVDD_SST	NAVDD_SST = 12h[6:4] = 0x00 to 0x07	2	--	16	ms
Default Power-on Soft-start Time	NAVDD_SST_4 MS	NAVDD_SST = 12h[6:4] = 0x01	3.6	4	4.4	ms
Power on Delay Time Range	NAVDD_DLY	NAVDD_DLY=12h[3:0] = 0x00 to 0x0F	0	--	45	ms
Default Power on Delay Time	NAVDD_DLY_1 0MS	NAVDD_DLY=12h[3:0] = 0x02	9	10	11	ms
VGH/VGL SIBO Converter						
VGH Output Voltage Range	VGH	VGH High Resolution 03h[7] = 0, VGH = 07h[4:0] = 0x00~0x14	10	--	30	V
		VGH High Resolution 03h[7] = 1, VGH = 07h[4:0] = 0x00~0x14	6	--	10	V
Default VGH Output Voltage	VGH_18V	VGH High Resolution 03h[7] = 0, VGH = 07h[4:0] = 0x08 = 18V	17.46	18	18.54	V
Default VGH Output Voltage	VGH_7.6V	VGH High Resolution 03h[7] = 1, VGH = 07h[4:0] = 0x08 = 7.6V	7.37	7.6	7.83	V

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
VGH Output Accuracy	VGH/L_ACC	VGH=10~30V/6V to 10V	-3	--	3	%
VGL Output Voltage Range	VGL	VGL = 08h[5:0] = 0x00~0x3F	-5.4	--	-18	V
Default VGL Output Voltage	VGL_-14V	VGL = 08h[5:0] = 0x00~0x2B = -14V	-14.42	-14	-13.58	V
VGL Output Accuracy	VGH/L_ACC	VGL = -5.4 to -18V	-3	--	3	%
Switch Frequency Range	VGH/L_FSW	VGH/L_FSW = 14h[2:0] = 0x00 to 0x05 = 600k / 715k / 800k / 900k / 1000k / 1200k	600	--	1200	kHz
Default Setting Switching Frequency	VGH/L_FSW_1200k	VGH/L_FSW = 14h[2:0] = 0x05 = 1200k	1080	1200	1320	kHz
Switch Frequency Accuracy	VGH/L_FSW_A CC_715k	VGH/L_FSW = 14h[2:0] = 0x01 = 715k	-5	--	5	%
	VGH/L_FSW_A CC	VGH/L_FSW = 600k / 800k / 900k / 1000k / 1200k	-10	--	10	%
Maximum Duty	VGH/L_DMAX		--	94	--	%
Maximum VGH/L Output Current Capability	IGH/L_MAX	VIN = 3V, VGH = 18V, VGL = -14V	20	--	--	mA
VGH/L Line Regulation	VGH/L_LIR	VIN = 2.5V to 5V	-0.3	--	0.3	%/V
VGH/L Load Regulation	VGH/L_LOR	VIN = 3.3V, I _{VGH/L} < 20mA	--	0.8	--	mV/mA
VGL High-Side MOSFET On-Resistance	VGL_RON_HS	IGH = 10mA	--	0.5	--	Ω
VGH Low-Side MOSFET On-Resistance	VGH_RON_LS	IGL = 10mA	--	0.5	--	Ω
LXN Leakage Current	VGH/L_LXN_ILK	VGL = -14V, VIN = 3.3V	--	--	1	μA
LXP Leakage Current	VGH/L_LXP_ILK	VLXP = 18V	--	--	1	μA
Current Limit Peak	VGH/L_ILIM_PEAK	VIN = 2.8V to 5V.	--	1.2	--	A
VGH/L Under Voltage Protection Threshold	VGH/L_UVP		--	80	--	%
VGH/L Under Voltage Protection Delay Time	VGH/L_UVP_DELAY	Trigger base on UVP	--	60	--	ms
VGH Short Circuit Protection Threshold	VGH_SCP		--	1.3	--	V
VGL Short Circuit Protection Threshold	VGL_SCP		--	20	--	%
VGH Power off Discharge Ron	VGH_RDIS	Before VGH soft-start and After UVLO_F should be turn on.	--	2.273	--	kΩ
VGH Power off Discharge Time	VGH_TDIS	Discharge function enable, C _{OUT} = 4.4μF, time = 100%~20%	--	30	--	ms
VGH Power-on Soft-start Time Range	VGH_SST	VGH_SST = 15h[5:4] = 0x00 to 0x07	4	--	12	ms
VGH Default Power-on Soft-start Time	VGH_SST_6MS	VGH_SST = 15h[5:4] = 0x01	5.4	6	6.6	ms

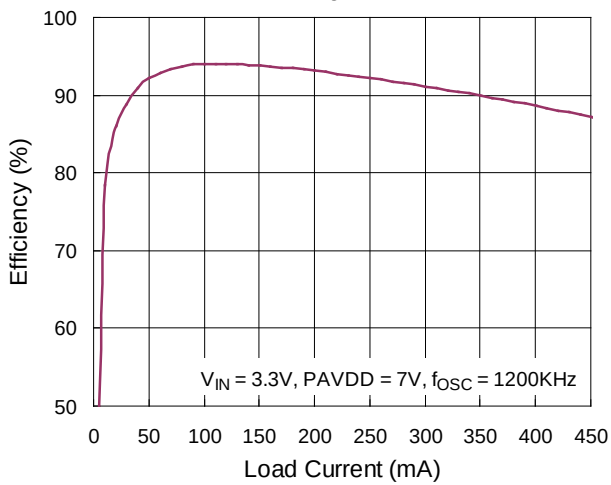
Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
VGH Power on Delay Time Range	VGH_DLY	VGH_DLY = 15h[3:0] = 0x00 to 0x0F	0	--	45	ms
VGH Default Power on Delay Time	VGH_DLY_25MS	VGH_DLY = 15h[3:0] = 0x05	22.5	25	27.5	ms
VGL Power off Discharge Ron	VGL_RDIS	Before VGL soft-start and After UVLO_F should be turn on.	--	0.75	--	kΩ
VGL Power off Discharge Time	VGL_TDIS	Discharge function enable, COUT = 4.4uF, time = 100% to 20%	--	10	--	ms
VGL Power-on Soft-start Time Range	VGL_SST	VGL_SST = 16h[5:4] = 0x00 to 0x03	4	--	12	ms
VGL Default Power-on Soft-start Time	VGL_SST_4MS	VGL_SST = 16h[5:4] = 0x00	3.6	4	4.4	ms
VGL Power on Delay Time Range	VGL_DLY	VGL_DLY = 16h[3:0] = 0x00 to 0x0F	0	--	45	ms
VGL Default Power on Delay Time	VGL_DLY_15MS	VGL_DLY = 16h[3:0] = 0x03	13.5	15	16.5	V
VCOM Operation Amplifier						
VCOM Output Voltage Range	VCOM	VCOM = 0Ch[7:0] = 0x00~0xFF	-2.56	--	1.06	V
Default VCOM Output Voltage	VCOM_-1V	VCOM = 0Ch[7:0] = 0x4E	-1.04	-1	-0.96	V
Digital VCOM Output Range	DVCOM	Slave = 0x9E, Data = 0x02~0xFF	VCOM - 10mV*64	--	VCOM + 10mV*64	V
Integral Non-Linearity	VCOM_INL	VCOM = -2.56V to 1.06V, No Load, Unit Gain, TA = 25°C	--	±1	--	LSB
Differential Non-Linearity	VCOM_DNL	VCOM = -2.56V to 1.06V, No Load, Unit Gain, TA = 25°C	--	±0.5	--	LSB
Output Source/Sink Current	ICOM_SI/O	ICOM = ±75mA, unit gain	--	±6	--	mV
Output Voltage Swing High	VCOM_VOH	ICOM = 100μA, VNEG = NAVDD	VIN - 0.03	VIN - 0.005	--	V
		ICOM = 75mA, VNEG = NAVDD	VIN - 1.9	VIN - 1.5	--	V
Output Voltage Swing Low	VCOM_VOL	ICOM = 100μA, VNEG = VIN	--	NAVDD + 0.005	NAVDD + 0.03	V
		ICOM=75mA, VNEG=VIN	--	NAVDD + 1.5	NAVDD + 1.9	V
NEG leakage current			--	--	0.1	μA
Output Short-Circuit Current	VCOM_ISC	Unit Gain, VCOM = -1V, Source = VCOM short to NAVDD/Sink = VCOM short to VIN	--	±285	--	mA
Power on/off Pull Down Resistance	VCOM_RPD		--	130	--	Ω
Power off Discharge Time	VCOM_TDIS	COUT = 4.7uF, time = 100% to 20%, Discharge function enable(Change Code)	--	1	--	ms
Power on Delay Time Range	VCOM_DLY	VCOM_DLY = 0Dh[7:3] = 0x00 to 0x1F	0	--	155	ms
Default Power on Delay Time	VCOM_DLY_25MS	VCOM_DLY = 0Dh[7:3] = 0x05	22.5	25	27.5	ms

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
GAM1/2 Operation Amplifier						
GAM1 Output Voltage Range	VGAM1	VGAM1 = 0Eh[5:0] = 0x00~0x3F	PAVDD – 1.26V	--	PAVDD	V
Default GAM1 Output Voltage	VGAM1_PAVDD -0.2V	VGAM1 = 0Eh[5:0] = 0x0A	PAVDD – 0.16	PAVDD – 0.2	PAVDD – 0.24	V
GAM2 Output Voltage Range	VGAM2	VGAM2 = 0Fh[5:0] = 0x00~0x3F	NAVDD	--	NAVDD + 1.26V	V
Default GAM2 Output Voltage	VGAM2_NAVDD +0.2V	VGAM2 = 0Fh[5:0] = 0x0A	NAVDD + 0.24	NAVDD + 0.2	NAVDD + 0.16	V
Integral Non-Linearity	GAM1/2_INL		--	±1	--	LSB
Differential Non-Linearity	GAM1/2_DNL		--	±0.5	--	LSB
Maximum Output Current Capability	IGAM1/2_MAX		5	--	--	mA
Dropout Voltage	GAM_DROP	PAVDD = 6V, VGAM1 = 5.8V, IGAM1/2 = 1mA	--	50	--	mV
Current Limit	GAM_ILIM		--	10	--	mA
Power on Delay Time Range	GAM1/2_DLY	GAM1/2_DLY = 0Eh[7:6] = 0x00 to 0x03	5	--	20	ms
Default Power on Delay Time	GAM1/2_DLY_1 0MS	GAM1/2_DLY = 0Eh[7:6] = 0x01	9	10	11	ms
GAM1 Output Voltage Swing High	GAM1_VOH	IGAM1 = 1mA	PAVDD – 0.2	--	--	V
GAM2 Output Voltage Swing Low	GAM2_VOL	IGAM2 = 1mA	--	--	NAVDD + 0.2	V
RESET Input Voltage Detector						
RESET Falling Voltage Range	VRESET	RESET = 0Dh[2:0] = 0x00~0x05	2.0	--	2.5	V
Default RESET Falling Voltage	VRESET_2.2V	RESET = 0Dh[2:0] = 0x02	2.15	2.2	2.25	V
RESET Detect Voltage Accuracy	VRESET_ACC		–2	--	2	%
RESET Rising Hysteresis Voltage	VRESET_HYS	VIN > UVLO	--	0.15	--	V
RESET Open-Drain On-Resistance	RESET_RON	Open-drain MOSFET is on, Vup = floating	--	100	--	Ω
RESET Open-Drain Leakage Current	RESET_ILK	Open-drain MOSFET is off, Vup = 3.3V	--	--	1	μA
Power-on Delay Time Range	RESET_DLY	RESET_DLY = 1Bh[3:0] = 0x00~0x0F	0	--	75	ms
Default Power-on Delay Time	RESET_DLY_15 MS	RESET_DLY = 1Bh[3:0] = 0x03	13.5	15	16.5	ms
EXT_EN						
EXT_EN Open-Drain On-Resistance	EXT_EN_RON	Open-drain MOSFET is on, Vup = floating	--	100	--	Ω
EXT_EN Open-Drain Leakage Current	EXT_EN_ILK	Open-drain is off, Vup = 3.3V	--	--	1	μA

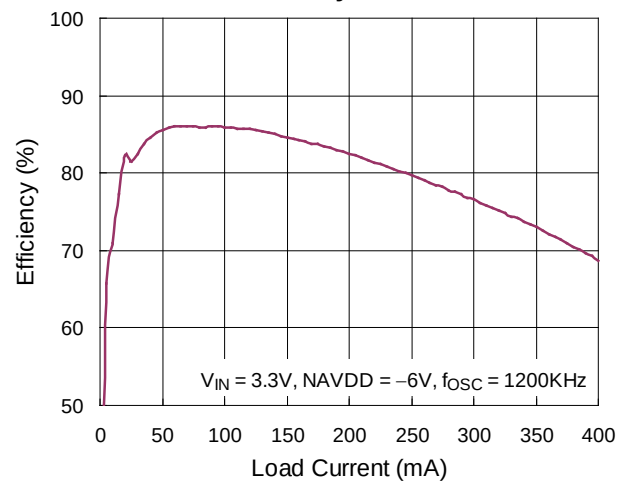
- Note 9.** Stresses beyond those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions may affect device reliability.
- Note 10.** θ_{JA} is measured under natural convection (still air) at $T_A = 25^\circ\text{C}$ with the component mounted on a high effective-thermal-conductivity four-layer test board on a JEDEC 51-7 thermal measurement standard. θ_{JC} is measured at the exposed pad of the package.
- Note 11.** Devices are ESD sensitive. Handling precaution is recommended.
- Note 12.** The device is not guaranteed to function outside its operating conditions.
- Note 13.** The spec is guaranteed to function outside its operating conditions.
- Note 14.** The SCL clock frequency operate at 1MHz, the minimum data set-up time is 50ns.

Typical Operating Characteristics

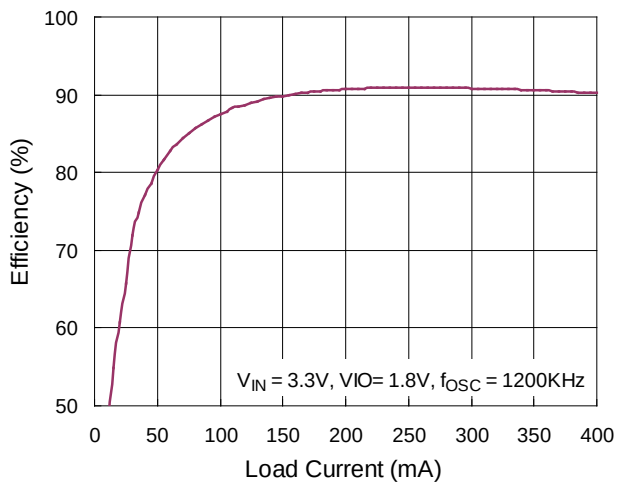
PAVDD Efficiency vs. Load Current



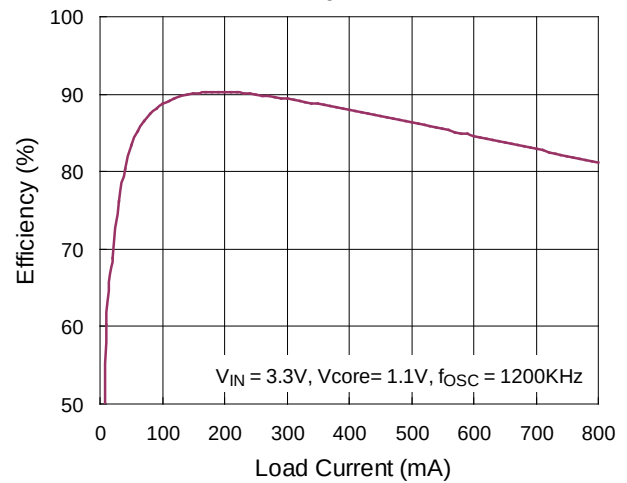
NAVDD Efficiency vs. Load Current



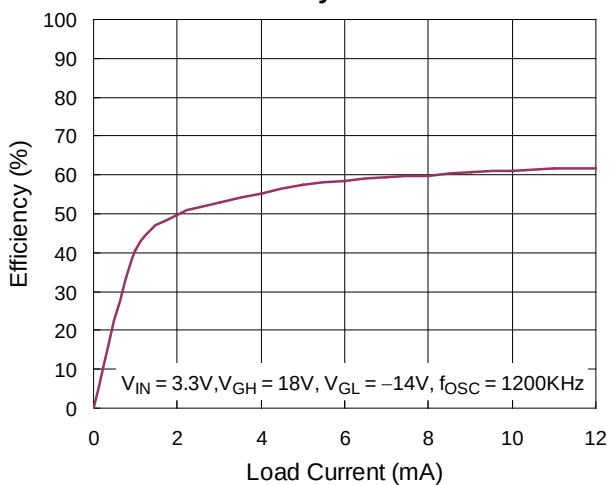
VIO Efficiency vs. Load Current



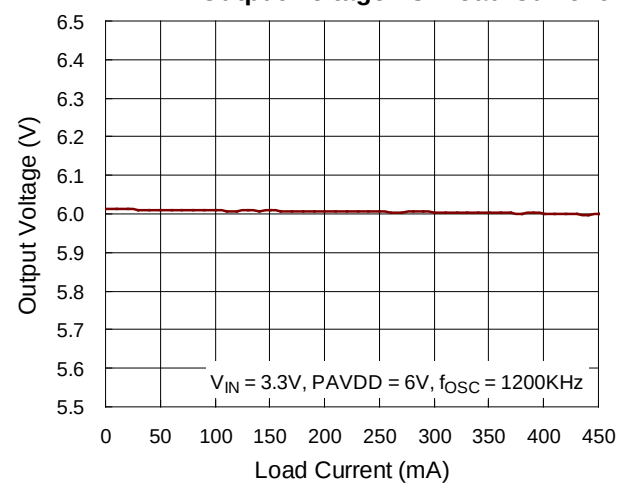
Vcore Efficiency vs. Load Current



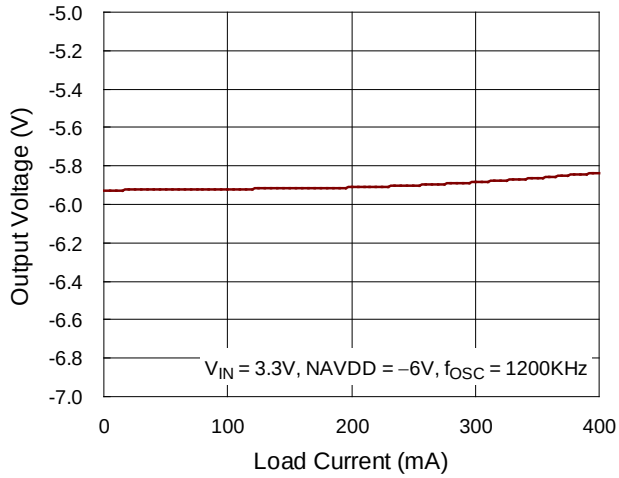
SIBO Efficiency vs. Load Current



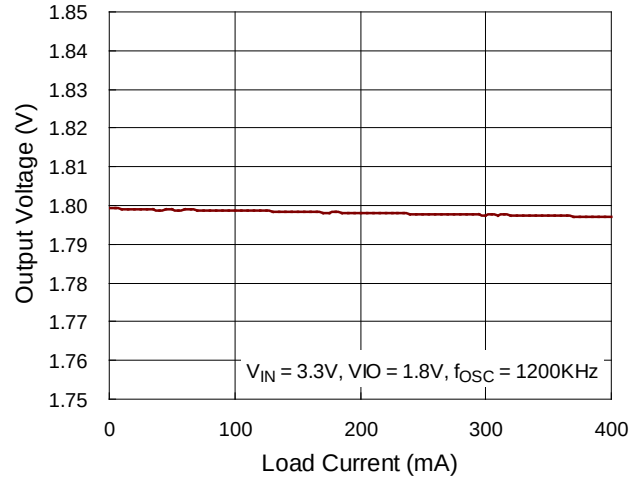
PAVDD Output Voltage vs. Load Current



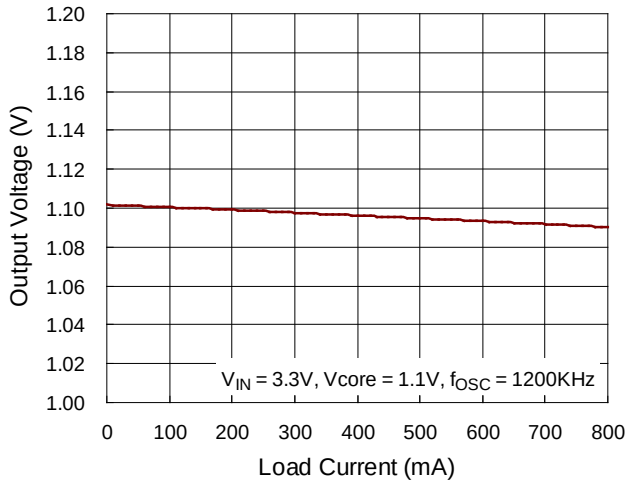
NAVDD Output Voltage vs. Load Current



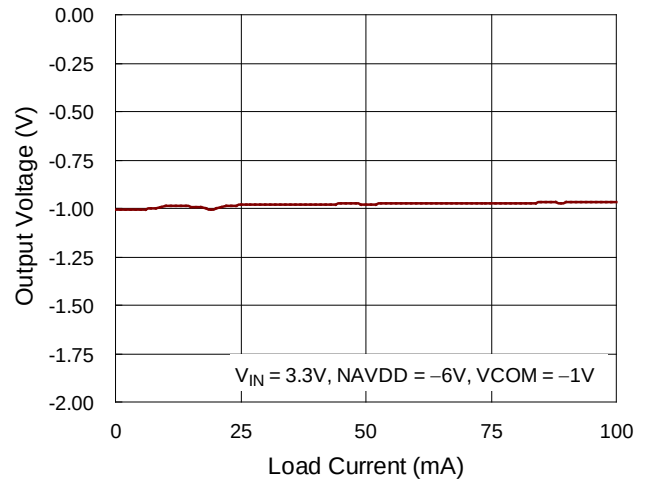
VIO Output Voltage vs. Load Current



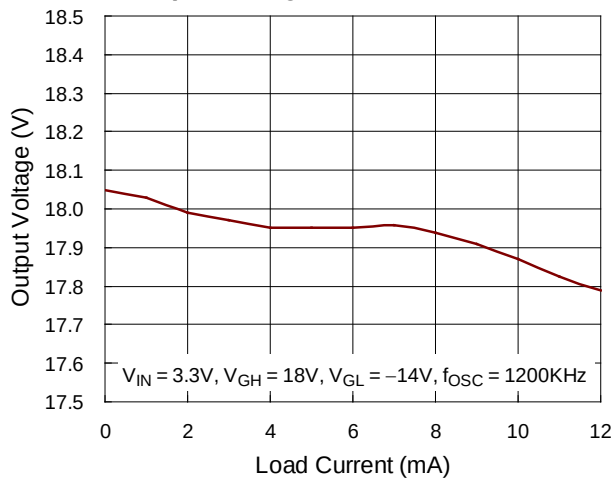
Vcore Output Voltage vs. Load Current



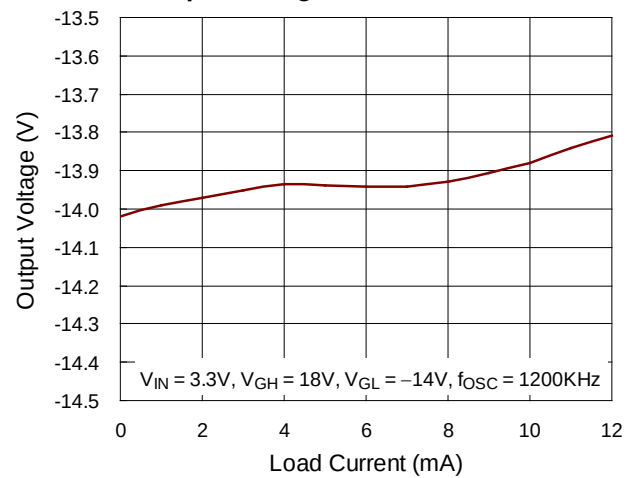
VCOM Output Voltage vs. Load Current



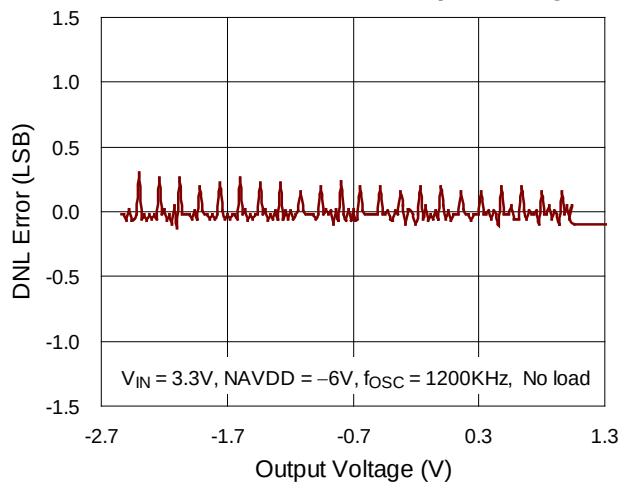
VGH Output Voltage vs. SIBO Load Current



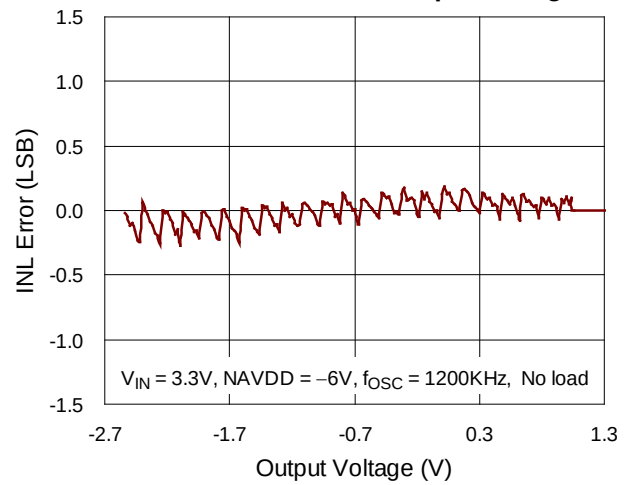
VGL Output Voltage vs. SIBO Load Current



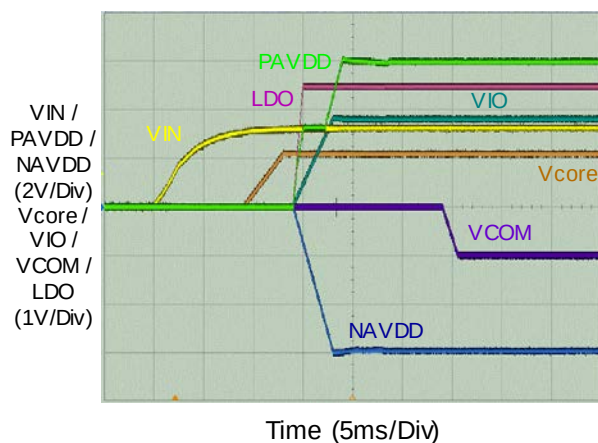
VCOM DNL Error vs. Output Voltage



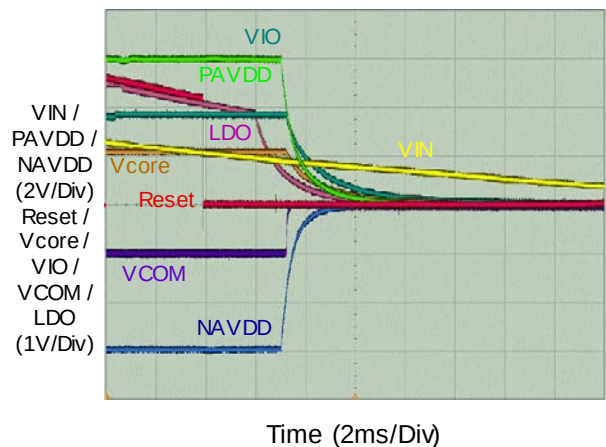
VCOM INL Error vs. Output Voltage



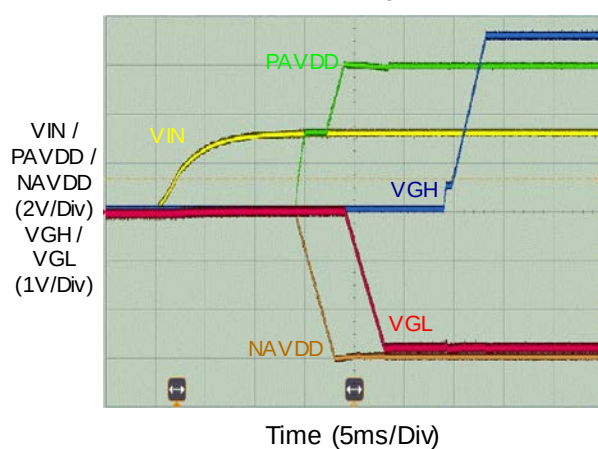
Power On Sequence I



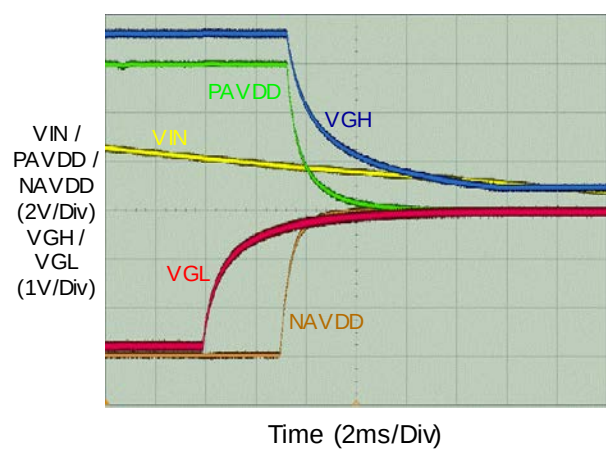
Power Off Sequence with Discharge I



Power On Sequence II



Power Off Sequence with Discharge II



Applications Information

The RT6755 is an integrated solution for TFT LCD panel, including PMIC and memory system. The RT6755 application mechanism is introduced in later sections. The RT6755's 7-bits slave address is 0100_011.

PMIC - Power management system provides a sync-boost converters for PAVDD, one synchronous inverting converter for NAVDD, a low dropout LDO, two synchronous Buck converters, a SIBO (Single Inductor Bipolar Output) converter for VGH and VGL, one digital VCOM operational amplifier with 7 bits calibrator, a high performance operation amplifier, 2-CH gamma, and a voltage detector. All channels of converters are programmable, such as output voltage, power on sequence, switching frequency of the converters, and to disable each output channel selection.

Channel Setting

The RT6755 can enable or disable each output by I2C interface. Users can control these outputs of PAVDD, NAVDD, VGH, VGL, VCORE, VIO, LDO, VCOM, GMA1, GMA2, and RESET, through programing the 02h and 03h register. The PAVDD, NAVDD, VGH, VGL, VCOM, NAVDD, VCORE, VIO, LDO, GMA1, GMA2 and RESET can be disable individually. Please refer the register map for details.

Under Voltage Protection (UVP)

The RT6755 equip a fault conditions to shut down IC. Once the output voltage is below the 80% output voltage, the internal timer starts counting and the fault condition continued about 60ms, the IC will shut down. The fault protection would be keep until VIN falls below UVLO. The protection start detecting at the soft-start finish of the outputs channel, the voltage of outputs should be larger than the UVP level after soft-start finish and within 60ms. The protection behavior is summarized in Table 2.

Short Circuit Protection (SCP)

The RT6755 equips SCP protection to shut down IC from damage.

For PAVDD and VGH, once output voltage falls below 1.3V, the converter will stop switching immediately. If

the fault condition does not release within 60ms after SCP occurred, all the other channels will be shut down and IC will latch.

For NAVDD, VIO, Vcore and VGL, once the output voltage is below 20% of target voltage, the corresponding converter will stop switching immediately. If the fault condition does not release within 60ms after SCP occurred, all the other channels will be shut down and latch.

The protection is released when UVLO is cycled. Please refer to the table 2 for the details.

Over Temperature Protection (OTP)

The RT6755 equips an over temperature protection (OTP) to prevent the excessive power dissipation from overheating. The OTP will shut down switching operation while junction temperature exceeds approximately 160°C. All of output channel starting work while junction temperature is cooled by approximately 25°C. Prevent the maximum junction temperature over around 150 °C and maintain continuous operation. Please refer to the table 2 for the details.

PAVDD Synchronous Boost Converter

The PAVDD synchronous Boost converter is high efficiency PWM architecture with programmable switching frequency. It performs fast transient responses to meet the requirement of source driver supplies for TFT-LCD display. The output voltage is controlled by a 6-bits register with 60 steps. The error amplifier varies the COMP voltage by sensing the PAVDD pin to regulate the output voltage.

PAVDD Switching Frequency Setting

The switching frequency is set by the I²C interface, the corresponding register is at 0x10h [2:0]. There are 5 steps in frequency setting for boost converter, the options are 600kHz, 715kHz, 800kHz, 900kHz, 1000kHz and 1200kHz. The switching frequency default value is 1200kHz (0x05h). Please refer to the register map for details.

PAVDD Slew Rate Setting

The PAVDD LX falling slew rate can be set by I²C interface, to optimize the efficiency and EMI performance. The corresponding register is at 0x10h[4:3]. The adjustable options are slowest(03h), slow(02h), normal(01h) and fast (00h). The default value is set as normal. Please refer to the register map for details.

PAVDD Output Voltage Setting

The PAVDD output voltage is set by I²C interface. User can program the 0x05h[5:0] register to set PAVDD output voltage. It has 6 bits for output voltage adjustable, the setting range is from 4V to 7V, and each voltage step is 50mV. The default voltage of PAVDD is 6V (0x28). Please refer the register map for detail on how to adjust the output voltage.

PAVDD Soft-start time Setting

The PAVDD soft-start time could be adjusted by the register 11h[6:4]. There are 3 bits and 8 steps. The soft-start time setting range is from 2ms to 16ms, and each step is 2ms. The soft-start time default value is 4ms (0x01). The soft-start sequence is following the reference voltage, the soft-start starting point is from the slope of the soft-start down to the point of crosses 0V. The soft-start finish point is PAVDD output voltage ready. Please refer to Figure 5 and register map for details.

PAVDD Power-on Delay Time Setting

The PAVDD power-on delay time is adjustable by I²C interface. There are 10 steps within 4 bits register of 11h. The delay time setting range is from 0ms to 45ms, and each steps time is about 5ms. The default value of delay time is 10ms (0x02). The delay time is from the MTP load data finish to PAVDD output voltage starting rising. Please refer the Figure 5, and register map for detail.

PAVDD Current Limit

The RT6755 can limit the peak current to achieve over-current protection. The IC senses the inductor current of on period that is flowing into LX pin. The minimum value of the current limit is 1A. The internal

N-MOSFET will be turned off if the peak inductor current achieve current limitation level, so that the output current at current limit boundary is denoted as I_{OUT(CL)} and can be calculated as shown in the following equation :

$$I_{OUT(CL)} = \eta \times \frac{V_{IN}}{V_{OUT}} \times \left(I_{CL} - \frac{1}{2} \times \frac{V_{IN} \times (V_{OUT} - V_{IN})}{V_{OUT}} \times \frac{T_S}{L} \right)$$

where η is the efficiency of the PAVDD sync-boost converter, I_{CL} is the value of the current limit and T_S is the switching period.

PAVDD Loop Compensation

The voltage feedback loop can be compensated with an external compensation network consisted of R7 and C13. Choose R7 to set high frequency integrator gain for fast transient response and C13 to set the integrator zero to maintain stability. The recommended values are 75k Ω and 470pF for most applications.

Sync-Boost Inductor Selection

The inductance depends on the maximum input current. The inductor ripple current range is 20% to 40% of maximum input current that is a general rule. If 40% is selected as an example, the inductor ripple current can be calculated as following equation :

$$I_{IN(MAX)} = \frac{V_{OUT} \times I_{OUT(MAX)}}{\eta \times V_{IN}}$$

$$I_{RIPPLE} = 0.4 \times I_{IN(MAX)}$$

Where η is the efficiency of the synchronous boost converter, $I_{IN(MAX)}$ is the maximum input current and I_{RIPPLE} is the inductor ripple current. Beside, the input peak current can be calculated by maximum input current plus half of inductor ripple current shown as following equation :

$$I_{PEAK} = 1.2 \times I_{IN(MAX)}$$

Note that the saturated current of inductor must be greater than I_{PEAK} . The inductance can be eventually determined as following equation :

$$L = \frac{\eta \times (V_{IN})^2 \times (V_{OUT} - V_{IN})}{0.4 \times (V_{OUT})^2 \times I_{OUT(MAX)} \times f_{OSC}}$$

Where f_{osc} is the PAVDD switching frequency. For better system performance, a shielded inductor is preferred to avoid EMI problems.

Sync-Boost Output Capacitor Selection

Output ripple voltage is an important index for estimating the performance. This portion consists of two parts, one is the product of $(I_{IN} + \frac{1}{2}\Delta I_L - I_{OUT})$ and ESR of output capacitor, another part is formed by charging and discharging process of output capacitor. Refer to Figure 6, evaluate ΔV_{OUT1} by ideal energy equalization. According to the definition of Q, the Q value can be calculated as following equation :

$$Q = \frac{1}{2} \times \left[\left(I_{IN} + \frac{1}{2}\Delta I_L - I_{OUT} \right) + \left(I_{IN} - \frac{1}{2}\Delta I_L - I_{OUT} \right) \right] \\ \times \frac{V_{IN}}{V_{OUT}} \times \frac{1}{f_{OSC}} = C_{OUT} \times \Delta V_{OUT1}$$

Where T_S is the inverse of switching frequency and the ΔI_L is the inductor ripple current. Move C_{OUT} to left side to estimate the value of ΔV_{OUT1} as following equation :

$$\Delta V_{OUT1} = \frac{D \times I_{OUT}}{\eta \times C_{OUT} \times f_{OSC}}$$

Then take the ESR into consideration, the ESR voltage can be determined as the following equation :

$$\Delta V_{ESR} = \left(\frac{I_{OUT}}{1-D} + \frac{V_{IN} \times D \times T_{OSC}}{2L} \right) \times R_{ESR}$$

Finally, the output ripple voltage ΔV_{OUT} is combined from the ΔV_{OUT1} and ΔV_{ESR} as following equation :

$$\Delta V_{OUT} = \Delta V_{OUT1} + \Delta V_{ESR}$$

In the general application, the PAVDD output capacitor is recommended that to use four 10 μ F/X5R/0603 capacitors.

NAVDD Synchronous Buck-Boost Converter

The NAVDD synchronous Buck-Boost converter is high efficiency PWM architecture with programmable switching frequency. It performs fast transient responses to meet the requirement of source driver supplies for TFT-LCD display. There are 31 steps

within 6-bits register to control output voltage.

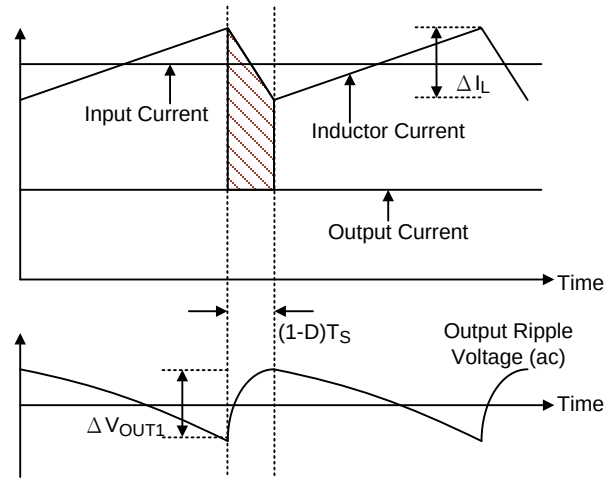


Figure 6. The Output Ripple Voltage without the Contribution of ESR

NAVDD Power-on Delay Time Setting

The NAVDD power-on delay time is adjustable by I2C interface. There are 10 steps within 4 bits register of 12h[3:0]. The power-on delay time setting range is from 0 ms to 45 ms, and each steps time is 5ms. The delay time default value is 10ms (0x02). The delay time is from the MTP load data finish to NAVDD output voltage starting falling. Please refer to register map for detail.

NAVDD Soft-Start Time Setting

The NAVDD has an internal soft-start mechanism to reduce the input inrush current. The NAVDD soft-start time can be adjusted by the register 12h[6:4]. There are 3 bits and 8 steps for setting. The soft-start time setting range is from 2 ms to 16 ms, and each step is 2 ms. The soft-start time default value is 4 ms (0x01). The soft-start time starts from the NAVDD delay time counting finish. The stop point of soft-start time is NAVDD output voltage ready. Please refer to Figure 5 and register map for details.

NAVDD Output Voltage Setting

The NAVDD output voltage is adjusted by I2C interface. User can write data into the register 06h[4:0]. There are 5 bits for output voltage adjustable, the setting range is from -4V to -7V, and each voltage step is -0.1V. The

default value is -6V(0x14). Please refer the register map for detail on how to adjust the output voltage.

NAVDD Inductor Selection

The NAVDD Buck-Boost converter is able to operate with 10 μ H inductor. The main parameter for inductor selection is the saturation current of the inductor which should be higher than the peak switch current as calculated in the Design Procedure section with additional margin to cover for heavy load transients. Another important parameter is the inductor DC resistance. Usually, lower DC resistance has higher efficiency. The type and core material of the inductor influence the efficiency as well.

NAVDD Output Capacitor Selection

For the best output voltage filtering, low ESR ceramic capacitors are recommended. Four 10 μ F/X5R/0603 capacitors in parallel are afford most applications. Additional capacitors can be added to improve output voltage ripple.

NAVDD Current Limitation

The RT6755 can limit the peak current to achieve over current protection. The IC senses the inductor current during an on period. The internal P-MOSFET will be turned off if the peak inductor current reaches achieve current limitation level. The NAVDD current limit can be adjusted by I2C interface, the corresponding register is at 0x13h [6:5]. The default value of NAVDD current limit is 1.5A (01h).

NAVDD Switching Frequency Setting

The switching frequency is set by the I²C interface, the corresponding register is at 0x13h[2:0]. There are 5 steps in frequency setting for boost converter, the options are 600kHz, 715kHz, 800kHz, 900kHz, 1000kHz and 1200kHz. The switching frequency default value is 1200kHz (0x05h). Please refer to the register map for details.

NAVDD Slew Rate Setting

The NAVDD LX falling slew rate can be set by I²C interface, to optimize the efficiency and EMI performance. The corresponding register is at

0x13h[4:3]. The adjustable options are slowest(03h), slow(02h), normal(01h) and fast (00h). The default value is set as normal. Please refer to the register map for details.

VIO and Vcore : Sync. Buck converter

The VIO and Vcore are sync-buck converters which are high efficiency PWM architecture with high switching frequency and fast transient response. The output voltage, switching frequency, LXB1 & LXB2 slew rate, soft-start time and delay time setting all can be controlled by the I²C interface.

VCORE Output Voltage setting

The VCORE sync-buck output voltage is set by the I²C interface. Users can program the 09h register to set output voltage. It has a 6-bits register for output voltage adjustment, the setting range is from 0.8V to 2V, and each voltage step is 20mV. The default output voltage of Vcore is 1.1V (0xFh). Please refer to the register map for details.

VCORE Delay Time Setting

The VCORE sync-buck converter has integrated a delay function. The delay time is adjustable through the 18h[5:4] register by the I²C interface. It has a 4-bits register with 10 steps. The delay time setting range is from 0ms to 45ms, and each step change is 5ms. The delay time default value is 5ms (0x01h). The delay time is defined as the VIN rises over the UVLO to output voltage of Vcore starts rising. Please refer to the Timing Diagram, and register map for details.

VCORE Soft-start Time Setting

The VCORE sync-buck converter has an integrated soft-start function to reduce the input inrush current of power on. The soft-start time is setting through the 18h[5:4] register by the I²C interface. It has a 2-bit register with 4 steps. The soft-start time setting range is from 2ms to 8ms, and each step change is 2ms. The soft-start time default value is 4ms (0x01h). The soft-start time is defined as the Vcore output voltage start rising to Vcore output voltage ready. Please refer to the Timing Diagram, and register map for details.

VCORE Slew Rate Setting

The VCORE sync-buck converter has a LXB1 slew rate control function to optimize the efficiency and EMI performance. It has a 2bits register with 4 steps, user can write the 17h[4:3] register to set LXB1 slew rate. There are four options for slew rate adjustment, fast, normal, slow, slowest. Set the register to 0x00h is fast. On the contrary, set 0x03h is slowest slew rate. The slew rate default value is 0x01h. Please refer to the register map for details.

VCORE Switching Frequency Setting

The switching frequency of sync-buck converter is set by I²C interface. It has a 3-bit register with 6 steps. The switching frequency range is from 600kHz to 1200kHz. The switching frequency default value is 1200kHz (0x05h). Please refer to the register map for details.

VIO Output Voltage Setting

The VIO sync-buck output voltage is set by the I²C interface. Users can write the 0Ah register to set sync-buck output voltage. It has a 5-bit register for output voltage adjustment, the setting range is from 1V to 2.5V, and each voltage step is 50mV. The default output voltage of VIO is 1.8V (0x10h). Please refer to the register map for details.

VIO Delay Time Setting

The VIO sync-buck converter has integrated a delay function. The delay time is adjustable by the I²C interface. It has a 4-bit register with 10 steps. The delay time setting range is from 0ms to 45ms, and each step change is 5ms. The delay time default value is 10ms (0x02h). The delay time is defined from the VIN rises over the UVLO to VIO output voltage starts rising. Please refer to the Timing Diagram, and register map for details.

VIO Soft-Start Time Setting

The VIO sync-buck converter has an integrated soft-start function to reduce the input inrush current of power on. The soft-start time is setting through the 1Ah[5:4] register by the I²C interface. It has a 2-bit register with 4 steps. The soft-start time setting range is from 2ms to 8ms, and each step change is 2ms. The

soft-start time default value is 4ms (0x01h). The soft-start time is defined from the VIO output voltage start rising to VIO output voltage ready. Please refer to the Timing Diagram, and register map for details.

VIO Slew Rate Setting

The VIO sync-buck converter has a LXB2 slew rate control function to optimize the efficiency and EMI performance. It has a 2bits register with 4 steps, user can write the 19h[4:3] register to set LXB2 slew rate. There are four options for slew rate adjustment, fast, normal, slow, slowest. Set the register to 0x00h is fast. On the contrary, set 0x03h is slowest slew rate. The slew rate default setting of VIO is normal (0x01h). Please refer to the register map for details.

VIO Switching Frequency Setting

The switching frequency of VIOsync-buck converter is set by I²C interface. It has a 3-bit register with 6 steps. The switching frequency range is from 600kHz to 1200kHz. The switching frequency default value is 1200kHz (0x05h). Please refer to the register map for details.

Sync. Buck Inductor Selection

The inductor value and operating frequency determine the ripple current according to a specific input and output voltage. The ripple current, ΔI_L , will increase with higher VIN and decrease with higher inductance, as shown in the equation below:

$$\Delta I_L = \left(\frac{V_{OUT}}{f_{SW} \times L} \right) \times \left(1 - \frac{V_{OUT}}{V_{IN}} \right)$$

Having a lower ripple current reduces not only the ESR losses in the output capacitors but also the output voltage ripple. High frequency with small ripple current can achieve the highest efficiency operation. However, it requires a large inductor to achieve this goal. For the ripple current selection, the value of $I_{L(MAX)} / \Delta I_L = 0.4$ is a reasonable starting point. The largest ripple current occurs at the highest VIN. To guarantee that the ripple current stays below the specified maximum, the inductor value should be chosen according to the following:

$$L = \left(\frac{V_{OUT}}{f_{SW} \times \Delta I_L(MAX)} \right) \times \left(1 - \frac{V_{OUT}}{V_{IN(MAX)}} \right)$$

Sync. Buck Output Capacitor Selection

The selection of C_{OUT} is determined by the required ESR to minimize voltage ripple. Moreover, the amount of bulk capacitance is also a key for C_{OUT} selection to ensure that the control loop is stable. Loop stability can be checked by viewing the load transient response. The output ripple, V_{OUT} , is determined by:

$$\Delta V_{OUT} = \Delta I_L \times \left(ESR + \frac{1}{8 \times f_{SW} \times C_{OUT}} \right)$$

The output ripple will be the highest at the maximum input voltage since I_L increases with input voltage. Multiple capacitors placed in parallel may be needed to meet the ESR and RMS current handling requirement. Four 10uF/X5R/0603 capacitors are recommended for V_{CORE} and Two 10uF/X5R/0603 capacitors are recommended for V_{IO}.

Sync. Buck Input Capacitor Selection

The input capacitance, C_{IN} is needed to filter the trapezoidal current at the source of the high-side MOSFET. To prevent large ripple current, a low ESR input capacitor sized for the maximum RMS current should be used. The RMS current is given by:

$$I_{RMS} = I_{OUT(MAX)} \times \frac{V_{OUT}}{V_{IN}} \times \sqrt{\frac{V_{IN}}{V_{OUT}} - 1}$$

This formula has a maximum at $V_{IN} = 2V_{OUT}$, where $I_{RMS} = I_{OUT}/2$. This simple worst-case condition is commonly used for design because even significant deviations do not offer much relief. Choose a capacitor rated at a higher temperature than required. Several capacitors may also be paralleled to meet size or height requirements in the design. For the input capacitor, a 10μF/6.3V low ESR ceramic capacitor is recommended.

VGH/VGL SIBO Converter

The VGH/VGL SIBO converter provides a boost mode for the positive (VGH) output voltage and an inverter mode for the negative (VGL) output voltage. The output voltage, switching frequency, soft-start time and delay time setting all can be controlled by the I²C interface. For flexibility in applications, the SIBO converter input,

V_{INP}, can be connected to either input voltage from typical V_{IN} or PAVDD output.

VGH Output Voltage Setting

The VGH output voltages are set by the I²C interface. Users can write the 01h[7] and 07h[4:0] register to set VGH output voltage. The resolution of output voltage can be adjusted by programming register 01h[7]. It has a 5-bit register for output voltage adjustment, the setting range is from 10V to 30V if 01h[7]=0x00h, and each voltage step is 1V. The default output voltage of VGH is 18V (0x08h). On the contrary, the setting range is from 6V to 10V at 01h[7]=0x01h, and each voltage step is 200mV. Please refer to the register map for details.

VGL Output Voltage Setting

The VGL output voltages are set by the I²C interface. Users can write 08h[5:0] register to set VGL output voltage. It has a 6-bit register for output voltage adjustment, the setting range is from -5.4V to -18V. The default output voltage of VGL is -14V (0x2Bh). Please refer to the register map for details.

VGH/VGL Switching Frequency Setting

The switching frequency of SIBO converter is set by I²C interface. It has a 3-bits register with 6 steps. The switching frequency range is from 600kHz to 1200kHz. The switching frequency default value is 1200kHz (0x05h). Please refer to the register map for details.

VGH/VGL Delay Time Setting

The VGH/VGL SIBO converter has integrated a delay function. The delay time is adjustable by the I²C interface. Both VGH/VGL have a 4-bits register with 16 steps. The delay time setting range is from 0ms to 75ms, and each step change is 5ms. The VGL delay time default value is 15ms (16h[3:0]=0x03h), the VGH delay time default value is 25ms (15h[3:0]=0x05h). The delay time is defined from the V_{IN} rises over the UVLO to VGL (VGH) output voltage starts falling (rising). For most application, it's recommended that set VGH delay time larger than VGL delay time plus VGL soft start time. Please refer to the Timing Diagram, and register map for details.

VGH/VGL Soft-start Time Setting

The VGH/VGL SIBO converter has an integrated soft-start function to reduce the input inrush current when power on. It has a 2-bits register with 4 steps. The soft-start time setting range is from 4ms to 12ms. The VGL soft-start time default value is 4ms (16h[5:4]=0x00h). The VGH soft-start time default value is 4ms (15h[5:4]=0x00h). The soft-start time is defined from the VGL (VGH) output voltage start falling (Rising) to VGL (VGH) output voltage ready. Please refer to the Timing Diagram, and register map for details.

LDO Output voltage Setting

The LDO linear regulator can supply up to 100mA current with an input voltage of 3.3V. It uses an internal P-MOSFET as the pass device. The LDO output voltage setting can be achieved by setting I2C register (0Bh[3:0]) from 1.8V to 3V. It has 4bit with 13 steps. The LDO default value is 2.5V(0x07h).

LDO Delay time Setting

The LDO linear regulator has integrated a delay function. The delay time is adjustable by the I2C interface. It has a 4-bits register with 10 steps, user can write the 0Bh[7:4] register to set LDO delay time. The delay time setting range is from 0ms to 45ms, and each step change is 5ms. The delay time default value is 10ms (0x02h). The delay time is defined from the VIN rises over the UVLO to LDO output voltage starts rising. Please refer to the Timing Diagram, and register map for details.

Programmable VCOM

The RT6755 provides the ability to reduce the flicker of an LCD Panel by adjusting the VCOM voltage during production test and alignment. The output voltage is adjusted by the I²C interface. There are two registers to adjust the VCOM voltage, one is VCOM for coarse tune, and another one is D-VCOM with independent slave ID 0x9Eh for fine tune. It is suggested to connect a resistor 10 ohm between output pin and output capacitor for better stability.

In general application, VCOM should not be set 0V. If user don't want to use this channel, please disable it.

VCOM Power-On Delay Time Setting

The VCOM is integrated power-on sequence control. The delay time is adjustable by I²C interface, there are 5 bits within register 0Dh[7:3]. The delay time setting range is from 0ms to 155ms, and each step is 5ms. The default value is 25ms (0x05h). The delay time is defined from the VIN rises over the UVLO to VCOM output voltage starts falling (rising). Please refer the Figure 5, and register map for detail.

VCOM Voltage Setting

The VCOM voltage is adjusted by the register 0x0Ch[7:0], it is provided 8bits resolution and 182 steps for user setting. The setting range is from -2.56V to 1.06V, each step is 20mV. The default value is -1V (0x4Eh). Please refer the register map for detail.

D-VCOM Voltage Setting

The D-VCOM is used to fine tune VCOM voltage and is programmable by I²C interface as well, but has independent slave ID 0x9E that is different than VCOM. User can adjust it with "D-VCOM I²C Write Timing Sequence" section in I2C the following. The D-VCOM provides 128 steps within 7 bits resolution. The default value is equal to VCOM if set D-VCOM as 0x40. Adjusting DVCOM upward from code 0x40 (center) means increasing VCOM finely, each step is 10mV upward. On the contrary, adjusting DVCOM downward from code 0x40 (center) means decreasing VCOM finely, each step is 10mV downward. The setting range is from (VCOM + 0.63V) to (VCOM – 0.64V). Please refer the register map for detail.

GMA1/GMA2

The RT6755 provides 2-CH gamma correction under 6-bits resolution. The adjustment of the output is provided by the I²C interface. Users can write the 0Eh (0Fh) register to set GMA1 (GMA2) output voltage. The GMA1 setting range is from PAVDD to PAVDD-1.26V. The GMA2 setting range is from NAVDD+1.26V to NAVDD.

RESET Voltage Detector

The voltage detector monitors the VIN voltage to generate a RESET signal from RESET pin while VIN is lower than the detecting level and not latched. Both detecting level and power-on delay time could be set by I²C interface. The detecting level could be adjusted by the register (0x0Dh [2:0]), it provided 6 options within 3 bits. The delay time could be set by register (0x1B[3:0]), the setting range is from 0ms to 75ms, the each step is about 15ms. The delay time start from that VIN over UVLO threshold, the stop point is that RESET signal goes to high.

In addition, the voltage detector can be turned off via I²C interface 1Bh[5:4]. There are four options can be selected: always high, or after RESET function is triggered, keep low in 20ms, 40ms, 60ms, then return to high level. Please refer to the Timing Diagram, and register map for details.

Discharge Function

The PAVDD, NAVDD, VGH, VGL, VIO, Vcore and VCOM outputs voltage is integrated a discharge function. The each output voltage discharged from 100% to 20% rapidly, preventing phenomena such as residual image on the display at power-off. The discharge function also provided an option for user to enable or disable, the option can set by the register 04h[8:0] individually for each channel.

Control Register (FFH)

The RT6755 provides a register for user choosing that write/read data into MTP or register. User can set the MSB of the register FFH goes to high, it means the data is written into MTP. But writing data into register don't need to set the register FFH. In addition, reading data from MTP need to set the LSB of register FFH to high. On the contrary, reading data from register need to set LSB to low. Please refer the "I²C Write/Read Timing Sequence" for detail.

Table 2. Protection Behavior Versus Protections

Block	Protection	Conditions	Behavior	Recovery
IC	OTP	IC Junction Temp. > 160°C(typ)	IC shut down.	
	VIN OVP	VIN > 5.85V(typ)	IC shut down.	VIN < VIN_UVLO_F.
PAVDD	OCP	ILX Peak > OCP setting(10h[6:5])	Normal operating: Cycle by cycle current limit.	Peak value of ILX < OCP
	UVP	PAVDD < 80% (typ)	Counting 60ms then all channel shut down and latch	VIN < VIN_UVLO_F.
	SCP	PAVDD < 1.3V (typ)	LX stop switching, and then all channel shut down and latch after 60ms	VIN < VIN_UVLO_F.
	Pre-charge NG	(VIN - PAVDD) > 0.2V	LX stop switching.	VIN - PAVDD < 0.2V
	OVP	PAVDD > 7.5V (typ)	LX stop switching.	PAVDD < OVP
NAVDD	OCP	ILXI Peak > OCP setting(13h[6:5])	Normal operating: Cycle by cycle current limit.	Peak value of ILXI < OCP
	UVP	NAVDD < 80% (typ)	Counting 60ms then all channel shut down and latch	VIN < VIN_UVLO_F.

Block	Protection	Conditions	Behavior	Recovery
	SCP	NAVDD < 20% (typ)	LXI stop switching, and then all channel shut down and latch after 60ms	VIN < VIN_UVLO_F.
Vcore	OCP	ILXB1 Peak > 1.5A (typ)	Normal operating: Cycle by cycle current limit.	Peak value of ILXB1 < OCP
	UVP	VCORE < 80% (typ)	Counting 60ms then all channel shut down and latch	VIN < VIN_UVLO_F.
	SCP	VCORE < 20% (typ)	LXB1 stop switching, and then all channel shut down and latch after 60ms	VIN < VIN_UVLO_F.
VIO	OCP	ILXB2 Peak > 1.2A (typ)	Normal operating: Cycle by cycle current limit.	Peak value of ILXB2 < OCP
	UVP	VIO < 80% (typ)	Counting 60ms then all channel shut down and latch	VIN < VIN_UVLO_F.
	SCP	VCORE < 20% (typ)	LXB2 stop switching, and then all channel shut down and latch after 60ms	VIN < VIN_UVLO_F.
VGH/L	OCP	ILXH/N Peak > 1A (typ)	Normal operating: Cycle by cycle current limit.	Peak value of ILXH/N < OCP
	UVP	VGH < 80% (typ)	Counting 60ms then all channel shut down and latch	VIN < VIN_UVLO_F.
		VGL < 80% (typ)		
	SCP	VGH < 1.3V (typ)	LXH stop switching, and then all channel shut down and latch after 60ms	VIN < VIN_UVLO_F.
		VGL < 20% (typ)	LXN stop switching, and then all channel shut down and latch after 60ms	
VCOM	SCP	IVCOM Over Current Limit > $\pm 285\text{mA}$ (typ)	Normal operating: Current limit.	IVCOM < Current Limit
GAM1/2	SCP	IGAM1/2 Over Current Limit > $\pm 5\text{mA}$ (min)	Normal operating: Current limit.	IGAM1/2 < Current Limit

I²C Command

PMIC Slave Address

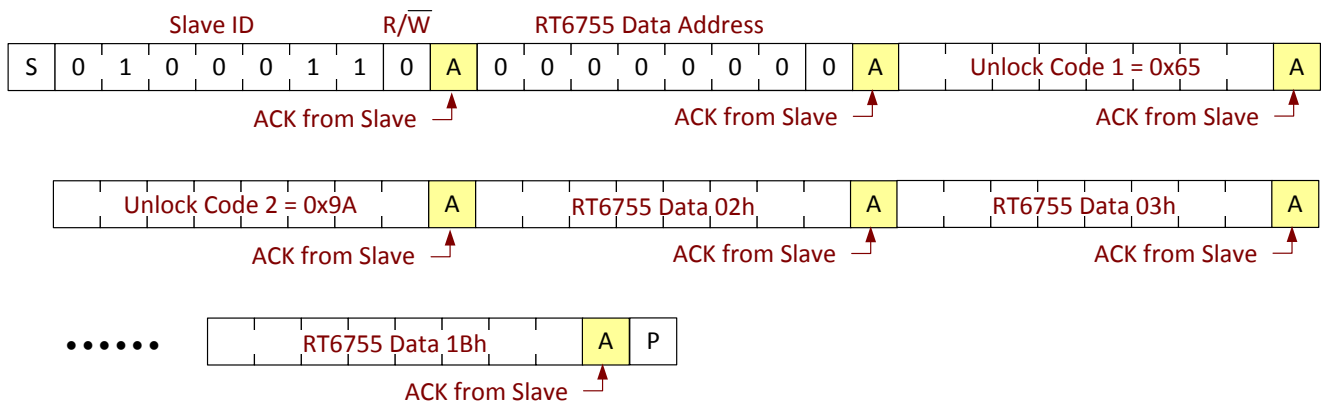
7	6	5	4	3	2	1	0=R/W	
0	1	0	0	0	1	1	0	46H
0	1	0	0	0	1	1	1	47H

D-VCOM Slave Address

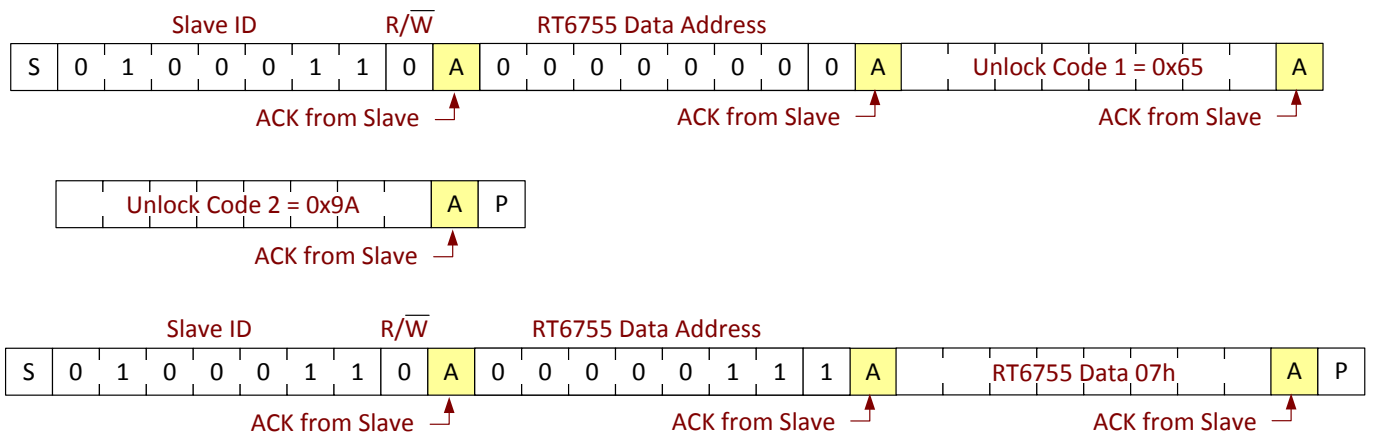
7	6	5	4	3	2	1	0=R/W	
1	0	0	1	1	1	1	0	9EH
1	1	1	1	1	1	1	1	9FH

PMIC I²C Write Timing Sequence (To DAC Register)

Write Multiple Data (00h~1Bh)

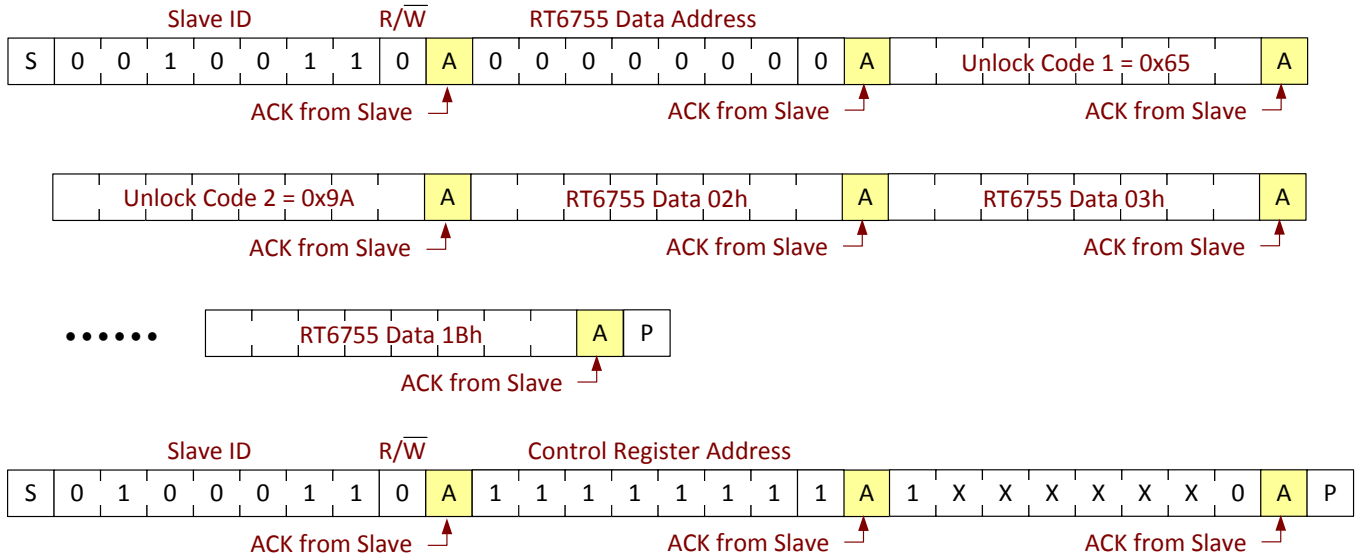


Write Single Data (07h)

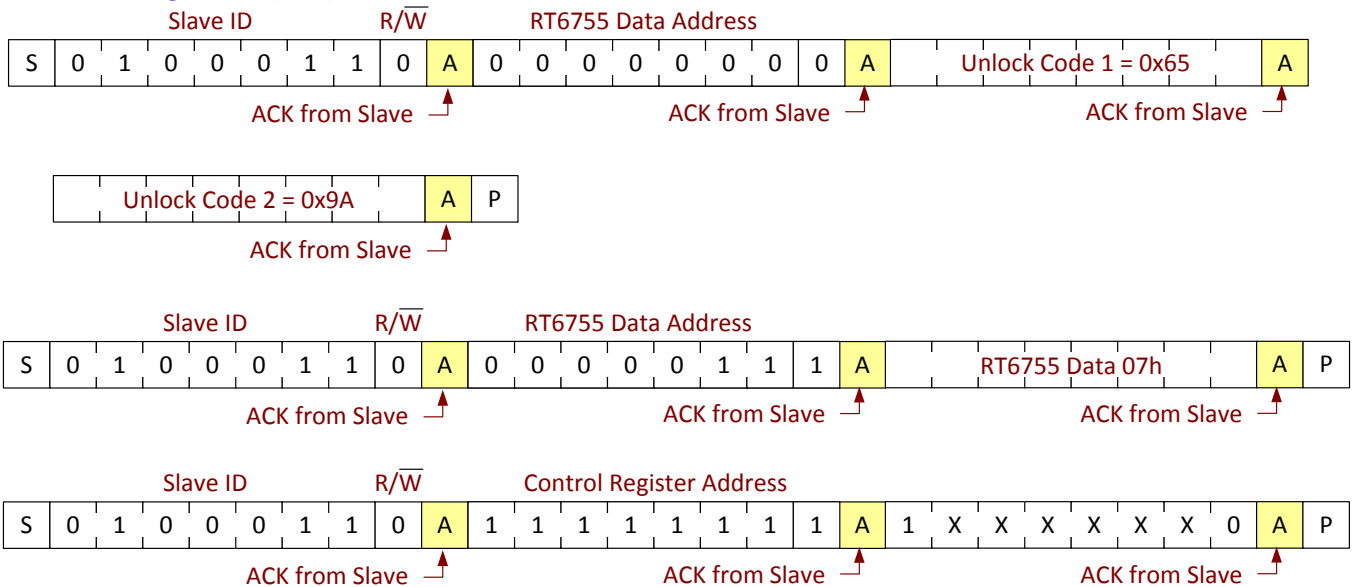


PMIC I²C Write Timing Sequence (To MTP)

Write Multiple Data (00h~1Bh)

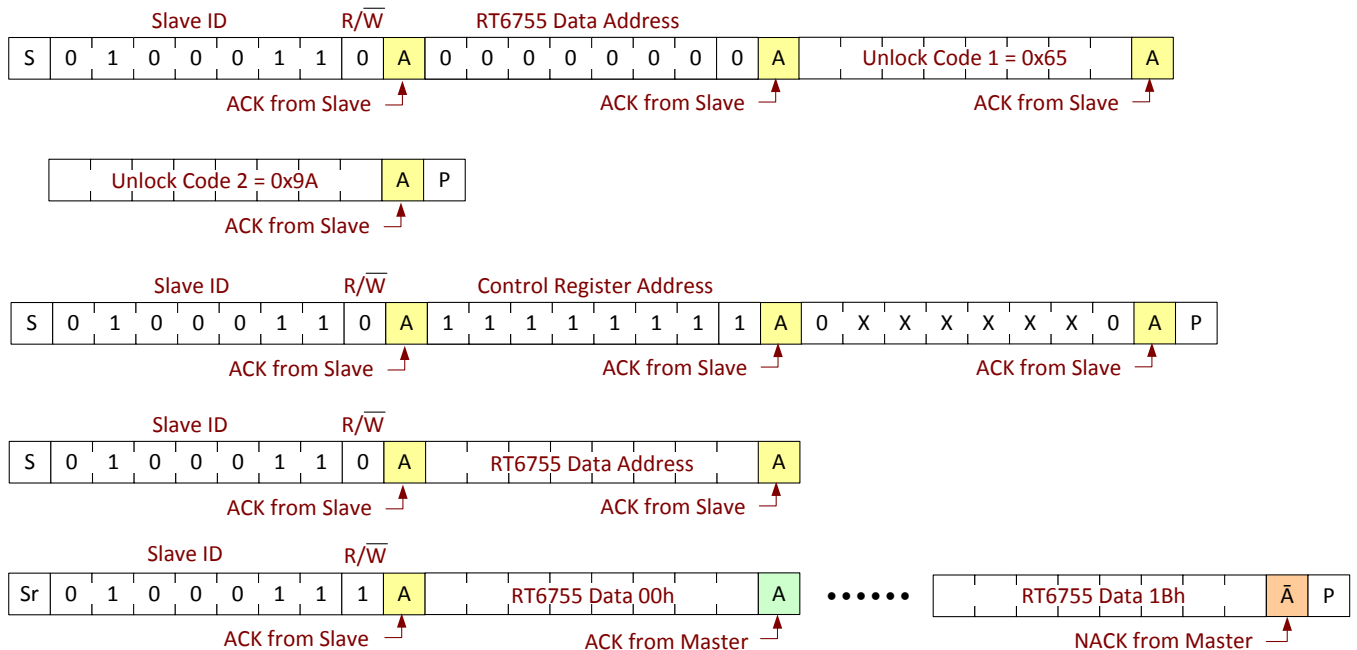


Write Single Data (07h)

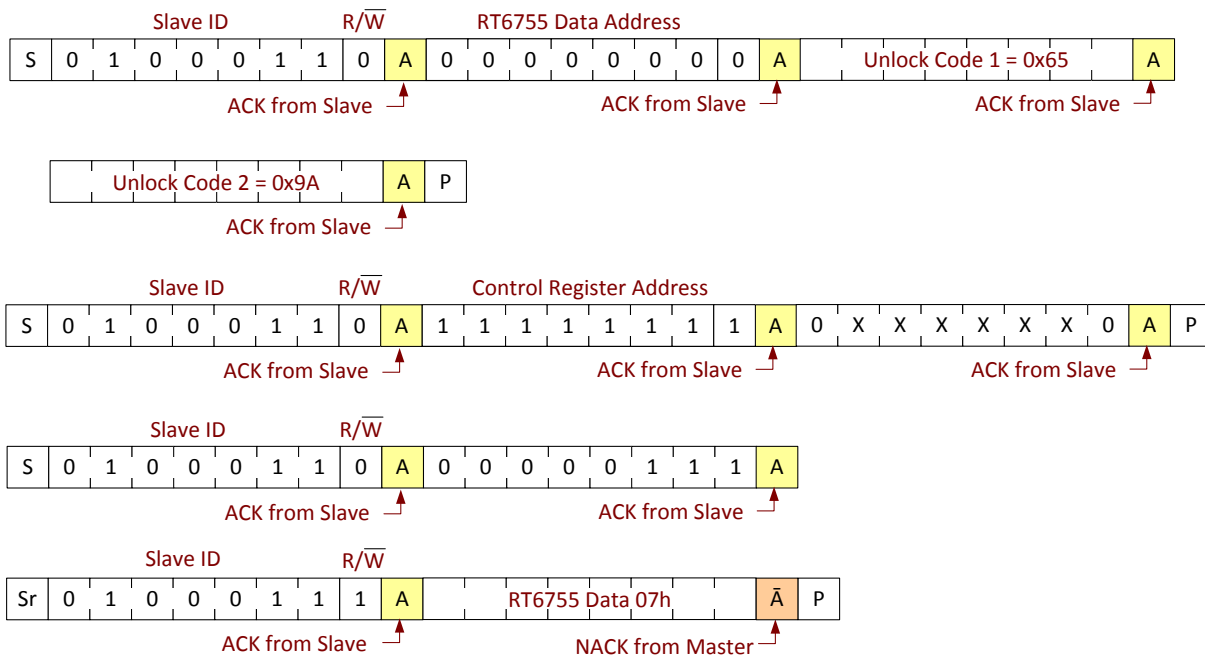


PMIC I²C Read Timing Sequence (From DAC Register)

Read Multiple Data (00h~1Bh)

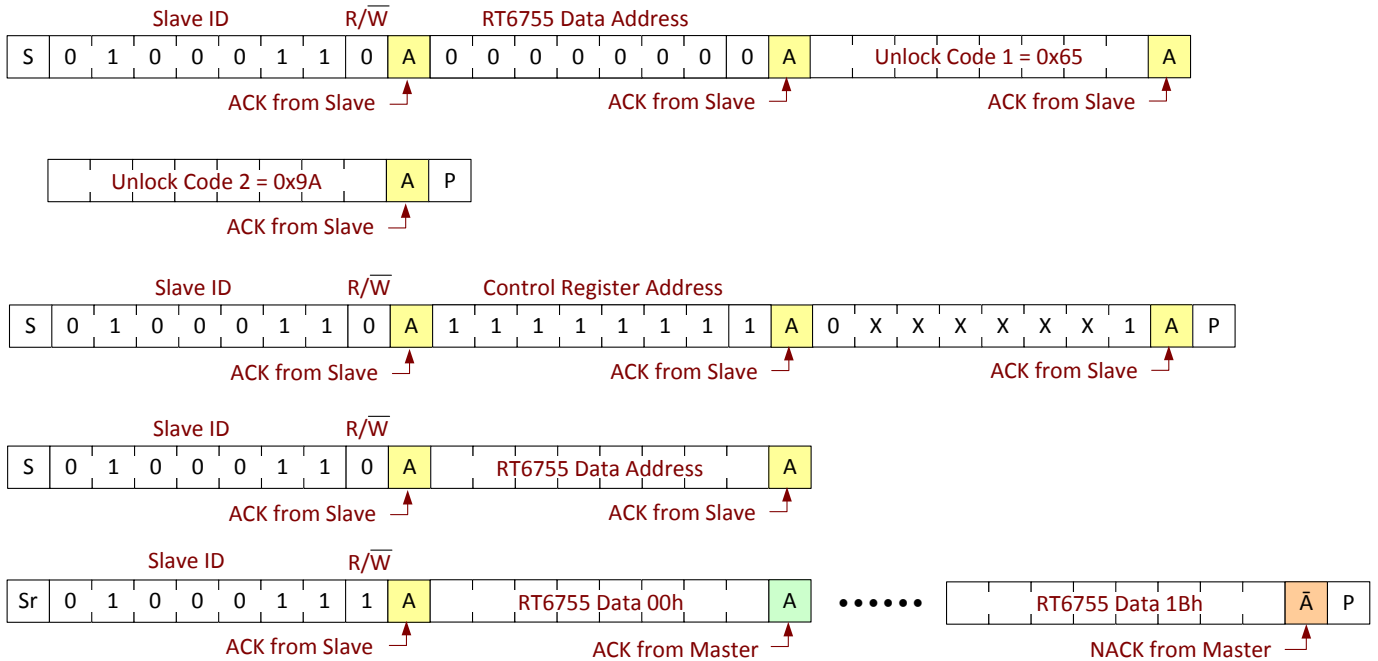


Read Single Data (07h)

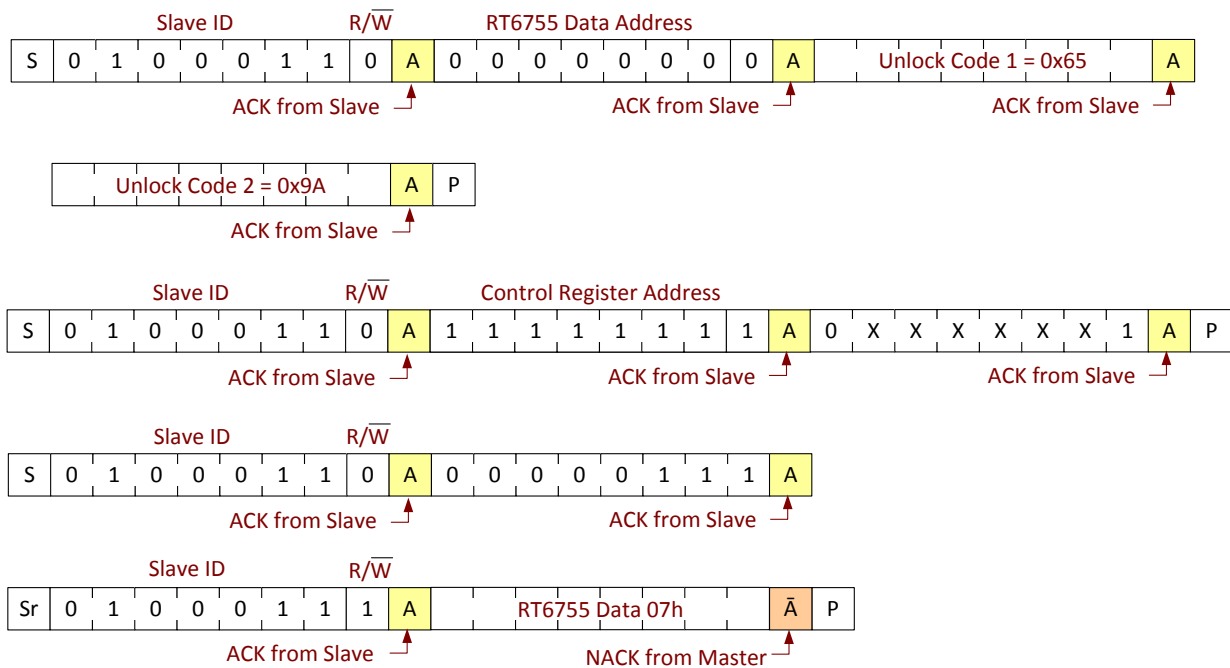


PMIC I²C Read Timing Sequence (From MTP)

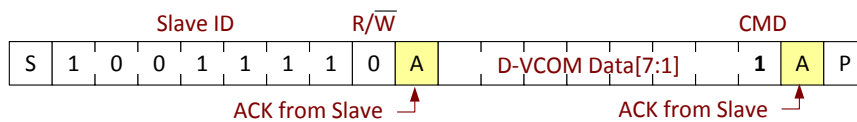
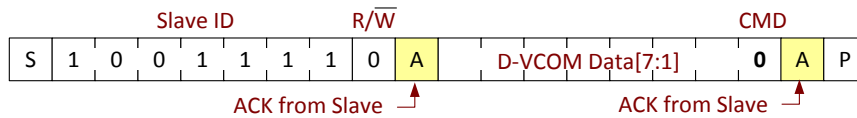
Read Multiple Data (00h~1Bh)



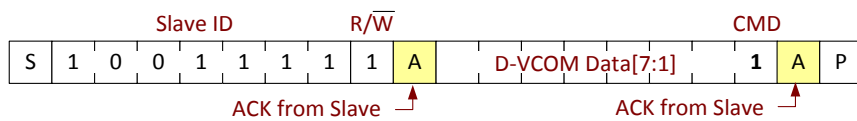
Read Single Data (07h)



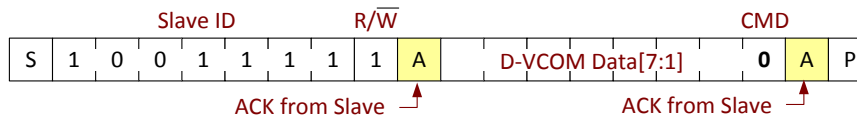
Note 15. If the I2C SCL frequency > 700kHz, only the single read can be used for MTP read.

D-VCOM I²C Write Timing Sequence (To DAC Register)**D-VCOM I²C Write Timing Sequence (To MTP)****D-VCOM I²C Read Timing Sequence (Data is the same as MTP)**

Read D-VCOM data from register(DAC) that register data is the same as EEPROM

**D-VCOM I²C Read Timing Sequence (Data is different to MTP)**

Read D-VCOM data from register(DAC) that register data is different to EEPROM



Register Map

Items		Register Address	Resolution	Range	Default value	Default Code	Step	Bit
WP	Soft-ware WP Code	00h	--	Unlock Code1=0x65	--	00h	--	8 Bit
		01h		Unlock Code2=0x9A		00h		8 Bit
Channel ON/OFF Setting	EN_VCOM[7]	02h	--	Disable(0)/Enable(1)	Enable	FFh	2 Steps	1 Bit
	EN_LDO[6]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_VIO[5]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_VCORE[4]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_VGL[3]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_VGH[2]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_NAVDD[1]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_PAVDD[0]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_VGH High Resolution(HR)[7]	03h	--	10V~30V(0)/6V~10V(1)	10V~30V	7h	2 Steps	1 Bit
	EN_VIO FCCM[6]		--	PSM(0)/FCCM(1)	PSM		2 Steps	1 Bit
	EN_VCORE FCCM[5]		--	PSM(0)/FCCM(1)	PSM		2 Steps	1 Bit
	EXT_EN[4:3]		--	Off, VCORE / NAVDD / PAVDD	Off		4 Steps	2 Bit
	EN_RESET[2]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_GMA2[1]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_GMA1[0]		--	Disable(0)/Enable(1)	Enable		2 Steps	1 Bit
	EN_VCOM Discharge [7]	04h	--	Disable(0)/Enable(1)	Disable	00h	2 Steps	1 Bit
Channel Discharge Setting	EN_LDO Discharge [6]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
	EN_VIO Discharge [5]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
	EN_VCORE Discharge [4]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
	EN_VGL Discharge [3]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
	EN_VGH Discharge [2]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
	EN_NAVDD Discharge [1]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
	EN_PAVDD Discharge [0]		--	Disable(0)/Enable(1)	Disable		2 Steps	1 Bit
PAVDD Voltage	PAVDD[5:0]	05h	50mV	4.0V~7V	6.0V	28h	60 Steps	6bit
NAVDD Voltage	Reserved[7:5]	06h	--	--	--	14h	--	3bit
	NAVDD[4:0]		-0.1V	-4.0V~-7V	-6.0V		30 Steps	5bit

Items		Register Address	Resolution	Range	Default value	Default Code	Step	Bit
VGH Voltage	Reserved[7:5]	07h	--	--	--	08h	--	3bit
	VGH[4:0] VGH_High_Res = 0		0.2V	6V~10V	7.6V		20 Steps	5bit
	VGH[4:0] VGH_High_Res = 1		1V	10V~30V	18V			5bit
VGL Voltage	Reserved[7:6]	08h	--	--	--	2Bh	--	2bit
	VGL[5:0]		-0.2V	-5.4V~-18V	-14V		63 Steps	6bit
VCORE Voltage	Reserved[7:6]	09h	--	--	--	0Fh	--	2bit
	VCORE[5:0]		20mV	0.8V~2.0V	1.1V		60 Steps	6bit
VIO Voltage	Reserved[7:5]	0Ah	--	--	--	10h	--	3bit
	VIO[4:0]		50mV	1V~2.5V	1.8V		30 Steps	5bit
LDO Delay	LDO Delay Time [7:4]	0Bh	5ms	0m~45m	10ms	27h	10 Steps	4bit
LDO Voltage	LDO[3:0]		0.1V	1.8V~3V	2.5V		12 Steps	4bit
VCOM Voltage	VCOM[7:0]	0Ch	20mV	-2.56V ~ 1.06V	-1.0V	4Eh	60 Steps	8bit
VCOM Delay	VCOM Delay Time [7:3]	0Dh	5ms	0m~155ms	25ms	2Ah	32 Steps	5bit
RESET Voltage	RESET[2:0]		0.1V	2.0~2.5V	2.2V		5 Steps	3bit
GMA1	GMA1[5:0]	0Eh	20mV	PAVDD ~ PAVDD-1.26V	PAVDD - 0.2V	0Ah	63 Steps	6bit
GMA2	GMA2[5:0]	0Fh	20mV	NAVDD ~ NAVDD+ 1.26V	NAVDD + 0.2V	0Ah	63 Steps	6bit
PAVDD Config	EN_PAVDD FCCM[7]	10h	--	PSM(0)/FCCM(1)	PSM	2Dh	2 Steps	1 Bit
	PAVDD Current Limit[6:5]		0.5A	1A~2.5A	1.5A		4 Steps	2bit
	PAVDD LX Slew Rate[4:3]		NA	Fast~ Slowest	Normal		4 Steps	2bit
	PAVDD LX Frequency[2:0]		NA	600~1200kHz	1200kHz		6 Steps	3bit
	PAVDD Soft-start Time[6:4]	11h	2ms	2m~16m	4ms	12h	8 Steps	3bit
	PAVDD Delay Time[3:0]		5ms	0m~45m	10ms		10 Steps	4bit
NAVDD Config	NAVDD Soft-start Time [6:4]	12h	2ms	2ms~16ms	4ms	12h	8 Steps	3bit
	NAVDD Delay Time [3:0]		5ms	0ms~45ms	10ms		10 Steps	4bit
	EN_NAVDD FCCM[7]	13h	--	PSM(0)/FCCM(1)	PSM	0Dh	2 Steps	1 Bit
	NAVDD Current Limit[6:5]		0.5A	1.5A~3A	1.5A		4 Steps	2bit
	NAVDD LXI Slew Rate[4:3]		NA	Fast~ Slowest	Normal		4 Steps	2bit
	NAVDD Frequency[2:0]		NA	600~1200kHz	1200kHz		6 Steps	3bit

Items		Register Address	Resolution	Range	Default value	Default Code	Step	Bit
VGH/VGL Config	LXH/N Slew Rate[4:3]	14h	NA	Fast~ Slowest	Normal	15h	4 Steps	2bit
	SIBO VINP Source[3]		NA	VIN / PAVDD	VIN		2 Steps	1bit
	LXP/LXN Frequency[2:0]		NA	600~1200kHz	1200kHz		6 Steps	3bit
	VGH Soft-start Time[5:4]	15h	NA	4m~12m	4ms	05h	4 Steps	2bit
	VGH Delay Time[3:0]		5ms	0m~75m	25ms		16 Steps	4bit
	VGL Soft-start Time[5:4]	16h	NA	4m~12m	4ms	03h	4 Steps	2bit
	VGL Delay Time[3:0]		5ms	0m~75m	15ms		16 Steps	4bit
VCORE Config	VCORE LXB1 Slew Rate[4:3]	17h	NA	Fast~ Slowest	Normal	0Dh	4 Steps	2bit
	VCORE LXB1 Frequency[2:0]		NA	600~1200kHz	1200kHz		6 Steps	3bit
	VCORE Soft-start Time[5:4]	18h	2ms	2m~8ms	4ms	11h	4 Steps	2bit
	VCORE Delay Time[3:0]		5ms	0m~45m	5ms		10 Steps	4bit
VIO Config	VIO LXB2 Slew Rate[4:3]	19h	NA	Fast~ Slowest	Normal	0Dh	4 Steps	2bit
	VIO LXB2 Frequency[2:0]		NA	600~1200kHz	1200kHz		6 Steps	3bit
	VIO Soft-start Time[5:4]	1Ah	2ms	2m~8ms	4ms	12h	4 Steps	2bit
	VIO Delay Time[3:0]		5ms	0m~45m	10ms		10 Steps	4bit
VCOM Config	EN_VCOM Pull to GND[7]	1Bh	NA	Disable(0)/Enable(1)	Enable	83h	2 Steps	1bit
	VCOM Power off Selection[6]		NA	Follow VIN UVLO_F(0) / Follow RESET(1)	Follow VIN UVLO_F		2 Steps	1bit
RESET Config	RESET Power off Control[5:4]		20ms	Off, 20ms/40ms/60ms	Off		4 Steps	2bit
	RESET Delay Time[3:0]		5ms	0m~75ms	15ms		16 Steps	4bit
Configuration Parameter VCOM	VCOM[7:1]	0x9E	10mV	VCOM-10mVx64~ VCOM+10mVx63	VCOM	40h	128 Steps	7bit
Control Register	00h : Read data form DAC	FFh						
	01h : Read data form EEPROM							
	80h : Write all DAC into EEPROM							

Register Table

	PAVDD Voltage	NAVDD Voltage		VGH Voltage			VGL Voltage	VCORE Voltage		VIO Voltage	
	PAVDD	Reserved	NAVDD	Reserved	VGH HR=1	VGH HR=0	VGL	Reserved	VCORE	Reserved	VIO
Data Address	05h	06h		07h			08h	09h		0Ah	
Bits	[5:0]	[7:5]	[4:0]	[7:5]	[4:0]	[4:0]	[5:0]	[7:6]	[5:0]	[7:5]	[4:0]
Min	4.0V	NA	-4.0V	NA	6V	10V	-5.4V	NA	0.8V	NA	1V
Max	7V	NA	-7V	NA	10V	30V	-18	NA	2.0V	NA	2.5V
Default	28h	00h	14h	00h	08h	08h	2Bh	00h	0Fh	00h	10h
Combination	28h	14h		08h			2Bh	0Fh		10h	
Resolution	0.05V	NA	-0.1V	NA	0.2V	1V	-0.2V	NA	0.02V	NA	0.05V
0H	4.00V	--	-4.0V	--	6.0V	10V	-5.4V	--	0.8V	--	1V
1H	4.05V	--	-4.1V	--	6.2V	11V	-5.6V	--	0.82V	--	1.05V
2H	4.10V	--	-4.2V	--	6.4V	12V	-5.8V	--	0.84V	--	1.10V
3H	4.15V	--	-4.3V	--	6.6V	13V	-6.0V	--	0.86V	--	1.15V
4H	4.20V	--	-4.4V	--	6.8V	14V	-6.2V	--	0.88V	--	1.20V
5H	4.25V	--	-4.5V	--	7.0V	15V	-6.4V	--	0.90V	--	1.25V
6H	4.30V	--	-4.6V	--	7.2V	16V	-6.6V	--	0.92V	--	1.30V
7H	4.35V	--	-4.7V	--	7.4V	17V	-6.8V	--	0.94V	--	1.35V
8H	4.40V	--	-4.8V	--	7.6V	18V	-7.0V	--	0.96V	--	1.40V
9H	4.45V	--	-4.9V	--	7.8V	19V	-7.2V	--	0.98V	--	1.45V
AH	4.50V	--	-5.0V	--	8.0V	20V	-7.4V	--	1.00V	--	1.50V
BH	4.55V	--	-5.1V	--	8.2V	21V	-7.6V	--	1.02V	--	1.55V
CH	4.60V	--	-5.2V	--	8.4V	22V	-7.8V	--	1.04V	--	1.60V
DH	4.65V	--	-5.3V	--	8.6V	23V	-8.0V	--	1.06V	--	1.65V
EH	4.70V	--	-5.4V	--	8.8V	24V	-8.2V	--	1.08V	--	1.70V
FH	4.75V	--	-5.5V	--	9.0V	25V	-8.4V	--	1.10V	--	1.75V
10H	4.80V	--	-5.6V	--	9.2V	26V	-8.6V	--	1.12V	--	1.80V
11H	4.85V	--	-5.7V	--	9.4V	27V	-8.8V	--	1.14V	--	1.85V
12H	4.90V	--	-5.8V	--	9.6V	28V	-9.0V	--	1.16V	--	1.90V
13H	4.95V	--	-5.9V	--	9.8V	29V	-9.2V	--	1.18V	--	1.95V
14H	5.00V	--	-6.0V	--	10.0V	30V	-9.4V	--	1.20V	--	2.00V
15H	5.05V	--	-6.1V	--			-9.6V	--	1.22V	--	2.05V
16H	5.10V	--	-6.2V	--			-9.8V	--	1.24V	--	2.10V
17H	5.15V	--	-6.3V	--			-10.0V	--	1.26V	--	2.15V
18H	5.20V	--	-6.4V	--			-10.2V	--	1.28V	--	2.20V
19H	5.25V	--	-6.5V	--			-10.4V	--	1.30V	--	2.25V
1AH	5.30V	--	-6.6V	--			-10.6V	--	1.32V	--	2.30V
1BH	5.35V	--	-6.7V	--			-10.8V	--	1.34V	--	2.35V
1CH	5.40V	--	-6.8V	--			-11.0V	--	1.36V	--	2.40V
1DH	5.45V	--	-6.9V	--			-11.2V	--	1.38V	--	2.45V
1EH	5.50V	--	-7.0V	--			-11.4V	--	1.40V	--	2.50V
1FH	5.55V						-11.6V		1.42V		
20H	5.60V						-11.8V		1.44V		
21H	5.65V						-12.0V		1.46V		
22H	5.70V						-12.2V		1.48V		
23H	5.75V						-12.4V		1.50V		
24H	5.80V						-12.6V		1.52V		
25H	5.85V						-12.8V		1.54V		
26H	5.90V						-13.0V		1.56V		
27H	5.95V						-13.2V		1.58V		

	PAVDD Voltage	NAVDD Voltage		VGH Voltage			VGL Voltage	VCORE Voltage		VIO Voltage	
	PAVDD	Reserved	NAVDD	Reserved	VGH HR=1	VGH HR=0	VGL	Reserved	VCORE	Reserved	VIO
28H	6.00V						-13.4V		1.60V		
29H	6.05V						-13.6V		1.62V		
2AH	6.10V						-13.8V		1.64V		
2BH	6.15V						-14.0V		1.66V		
2CH	6.20V						-14.2V		1.68V		
2DH	6.25V						-14.4V		1.70V		
2EH	6.30V						-14.6V		1.72V		
2FH	6.35V						-14.8V		1.74V		
30H	6.40V						-15.0V		1.76V		
31H	6.45V						-15.2V		1.78V		
32H	6.50V						-15.4V		1.80V		
33H	6.55V						-15.6V		1.82V		
34H	6.60V						-15.8V		1.84V		
35H	6.65V						-16.0V		1.86V		
36H	6.70V						-16.2V		1.88V		
37H	6.75V						-16.4V		1.90V		
38H	6.80V						-16.6V		1.92V		
39H	6.85V						-16.8V		1.94V		
3AH	6.90V						-17.0V		1.96V		
3BH	6.95V						-17.2V		1.98V		
3CH	7.00V						-17.4V		2.00V		
3DH							-17.6V				
3EH							-17.8V				
3FH							-18.0V				

	LDO Delay Time	LDO Voltage	VCOM Voltage	VCOM Delay Time	RESET Voltage	GMA1 Voltage	GMA2 Voltage
	LDO Delay Time	LDO	VCOM	VCOM Delay Time	RESET	GMA1	GMA2
Data Address	0Bh		0Ch	0Dh		0Eh	0Fh
Bits	[7:4]	[3:0]	[7:0]	[7:3]	[2:0]	[5:0]	[5:0]
Min	0ms	1.8V	-2.56V	0ms	2.0V	PAVDD-1.26V	NAVDD
Max	45ms	3.0V	1.06V	155ms	2.5V	PAVDD	NAVDD+1.26V
Default	2h	7h	4Eh	05h	02h	0Ah	0Ah
Combination	27h		4Eh	2Ah		0Ah	0Ah
Resolution	5ms	0.1V	0.02V	5ms	0.1V	20mV	20mV
0H	0ms	1.8V	-2.56V	0ms	2V	PAVDD	NAVDD
1H	5ms	1.9V	-2.54V	5ms	2.1V	PAVDD-0.02V	NAVDD+0.02V
2H	10ms	2.0V	-2.52V	10ms	2.2V	PAVDD-0.04V	NAVDD+0.04V
3H	15ms	2.1V	-2.50V	15ms	2.3V	PAVDD-0.06V	NAVDD+0.06V
4H	20ms	2.2V	-2.48V	20ms	2.4V	PAVDD-0.08V	NAVDD+0.08V
5H	25ms	2.3V	-2.46V	25ms	2.5V	PAVDD-0.10V	NAVDD+0.10V
6H	30ms	2.4V	-2.44V	30ms		PAVDD-0.12V	NAVDD+0.12V
7H	35ms	2.5V	-2.42V	35ms		PAVDD-0.14V	NAVDD+0.14V
8H	40ms	2.6V	-2.40V	40ms		PAVDD-0.16V	NAVDD+0.16V
9H	45ms	2.7V	-2.38V	45ms		PAVDD-0.18V	NAVDD+0.18V

	LDO Delay Time	LDO Voltage	VCOM Voltage	VCOM Delay Time	RESET Voltage	GMA1 Voltage	GMA2 Voltage
	LDO Delay Time	LDO	VCOM	VCOM Delay Time	RESET	GMA1	GMA2
AH		2.8V	-2.36V	50ms		PAVDD-0.20V	NAVDD+0.20V
BH		2.9V	-2.34V	55ms		PAVDD-0.22V	NAVDD+0.22V
CH		3.0V	-2.32V	60ms		PAVDD-0.24V	NAVDD+0.24V
DH			-2.30V	65ms		PAVDD-0.26V	NAVDD+0.26V
EH			-2.28V	70ms		PAVDD-0.28V	NAVDD+0.28V
FH			-2.26V	75ms		PAVDD-0.30V	NAVDD+0.30V
10H			-2.24V	80ms		PAVDD-0.32V	NAVDD+0.32V
11H			-2.22V	85ms		PAVDD-0.34V	NAVDD+0.34V
12H			-2.20V	90ms		PAVDD-0.36V	NAVDD+0.36V
13H			-2.18V	95ms		PAVDD-0.38V	NAVDD+0.38V
14H			-2.16V	100ms		PAVDD-0.40V	NAVDD+0.40V
15H			-2.14V	105ms		PAVDD-0.42V	NAVDD+0.42V
16H			-2.12V	110ms		PAVDD-0.44V	NAVDD+0.44V
17H			-2.10V	115ms		PAVDD-0.46V	NAVDD+0.46V
18H			-2.08V	120ms		PAVDD-0.48V	NAVDD+0.48V
19H			-2.06V	125ms		PAVDD-0.50V	NAVDD+0.50V
1AH			-2.04V	130ms		PAVDD-0.52V	NAVDD+0.52V
1BH			-2.02V	135ms		PAVDD-0.54V	NAVDD+0.54V
1CH			-2.00V	140ms		PAVDD-0.56V	NAVDD+0.56V
1DH			-1.98V	145ms		PAVDD-0.58V	NAVDD+0.58V
1EH			-1.96V	150ms		PAVDD-0.60V	NAVDD+0.60V
1FH			-1.94V	155ms		PAVDD-0.62V	NAVDD+0.62V
20H			-1.92V			PAVDD-0.64V	NAVDD+0.64V
21H			-1.90V			PAVDD-0.66V	NAVDD+0.66V
22H			-1.88V			PAVDD-0.68V	NAVDD+0.68V
23H			-1.86V			PAVDD-0.70V	NAVDD+0.70V
24H			-1.84V			PAVDD-0.72V	NAVDD+0.72V
25H			-1.82V			PAVDD-0.74V	NAVDD+0.74V
26H			-1.80V			PAVDD-0.76V	NAVDD+0.76V
27H			-1.78V			PAVDD-0.78V	NAVDD+0.78V
28H			-1.76V			PAVDD-0.80V	NAVDD+0.80V
29H			-1.74V			PAVDD-0.82V	NAVDD+0.82V
2AH			-1.72V			PAVDD-0.84V	NAVDD+0.84V
2BH			-1.70V			PAVDD-0.86V	NAVDD+0.86V
2CH			-1.68V			PAVDD-0.88V	NAVDD+0.88V
2DH			-1.66V			PAVDD-0.90V	NAVDD+0.90V
2EH			-1.64V			PAVDD-0.92V	NAVDD+0.92V
2FH			-1.62V			PAVDD-0.94V	NAVDD+0.94V
30H			-1.60V			PAVDD-0.96V	NAVDD+0.96V
31H			-1.58V			PAVDD-0.98V	NAVDD+0.98V
32H			-1.56V			PAVDD-1.00V	NAVDD+1.00V
33H			-1.54V			PAVDD-1.02V	NAVDD+1.02V
34H			-1.52V			PAVDD-1.04V	NAVDD+1.04V
35H			-1.50V			PAVDD-1.06V	NAVDD+1.06V
36H			-1.48V			PAVDD-1.08V	NAVDD+1.08V
37H			-1.46V			PAVDD-1.10V	NAVDD+1.10V
38H			-1.44V			PAVDD-1.12V	NAVDD+1.12V
39H			-1.42V			PAVDD-1.14V	NAVDD+1.14V
3AH			-1.40V			PAVDD-1.16V	NAVDD+1.16V

	LDO Delay Time	LDO Voltage	VCOM Voltage	VCOM Delay Time	RESET Voltage	GMA1 Voltage	GMA2 Voltage
	LDO Delay Time	LDO	VCOM	VCOM Delay Time	RESET	GMA1	GMA2
3BH			-1.38V			PAVDD-1.18V	NAVDD+1.18V
3CH			-1.36V			PAVDD-1.20V	NAVDD+1.20V
3DH			-1.34V			PAVDD-1.22V	NAVDD+1.22V
3EH			-1.32V			PAVDD-1.24V	NAVDD+1.24V
3FH			-1.30V			PAVDD-1.26V	NAVDD+1.26V
40H			-1.28V				
41H			-1.26V				
42H			-1.24V				
43H			-1.22V				
44H			-1.20V				
45H			-1.18V				
46H			-1.16V				
47H			-1.14V				
48H			-1.12V				
49H			-1.10V				
4AH			-1.08V				
4BH			-1.06V				
4CH			-1.04V				
4DH			-1.02V				
4EH			-1.00V				
4FH			-0.98V				
50H			-0.96V				
51H			-0.94V				
52H			-0.92V				
53H			-0.90V				
54H			-0.88V				
55H			-0.86V				
56H			-0.84V				
57H			-0.82V				
58H			-0.80V				
59H			-0.78V				
5AH			-0.76V				
5BH			-0.74V				
5CH			-0.72V				
5DH			-0.70V				
5EH			-0.68V				
5FH			-0.66V				
60H			-0.64V				
61H			-0.62V				
62H			-0.60V				
63H			-0.58V				
64H			-0.56V				
65H			-0.54V				
66H			-0.52V				
67H			-0.50V				
68H			-0.48V				
69H			-0.46V				
6AH			-0.44V				
6BH			-0.42V				

	LDO Delay Time	LDO Voltage	VCOM Voltage	VCOM Delay Time	RESET Voltage	GMA1 Voltage	GMA2 Voltage
	LDO Delay Time	LDO	VCOM	VCOM Delay Time	RESET	GMA1	GMA2
6CH			-0.40V				
6DH			-0.38V				
6EH			-0.36V				
6FH			-0.34V				
70H			-0.32V				
71H			-0.30V				
72H			-0.28V				
73H			-0.26V				
74H			-0.24V				
75H			-0.22V				
76H			-0.20V				
77H			-0.18V				
78H			-0.16V				
79H			-0.14V				
7AH			-0.12V				
7BH			-0.10V				
7CH			-0.08V				
7DH			-0.06V				
7EH			-0.04V				
7FH			-0.02V				
80H			0.00V				
81H			0.02V				
82H			0.04V				
83H			0.06V				
84H			0.08V				
85H			0.10V				
86H			0.12V				
87H			0.14V				
88H			0.16V				
89H			0.18V				
8AH			0.20V				
8BH			0.22V				
8CH			0.24V				
8DH			0.26V				
8EH			0.28V				
8FH			0.30V				
90H			0.32V				
91H			0.34V				
92H			0.36V				
93H			0.38V				
94H			0.40V				
95H			0.42V				
96H			0.44V				
97H			0.46V				
98H			0.48V				
99H			0.50V				
9AH			0.52V				
9BH			0.54V				
9CH			0.56V				

	LDO Delay Time	LDO Voltage	VCOM Voltage	VCOM Delay Time	RESET Voltage	GMA1 Voltage	GMA2 Voltage
	LDO Delay Time	LDO	VCOM	VCOM Delay Time	RESET	GMA1	GMA2
9DH			0.58V				
9EH			0.60V				
9FH			0.62V				
A0H			0.64V				
A1H			0.66V				
A2H			0.68V				
A3H			0.70V				
A4H			0.72V				
A5H			0.74V				
A6H			0.76V				
A7H			0.78V				
A8H			0.80V				
A9H			0.82V				
AAH			0.84V				
ABH			0.86V				
ACH			0.88V				
ADH			0.90V				
AEH			0.92V				
AFH			0.94V				
B0H			0.96V				
B1H			0.98V				
B2H			1.00V				
B3H			1.02V				
B4H			1.04V				
B5H			1.06V				

	PAVDD Config					
	EN_PAVDD FCCM	PAVDD Current Limit	PAVDD LX FSR	PAVDD LX Frequency	PAVDD Soft-start Time	PAVDD Delay Time
Data Address	10h				11h	
Bits	[7]	[6:5]	[4:3]	[2:0]	[6:4]	[3:0]
Min	NA	1A	Slow	600kHz	2ms	0ms
Max	NA	2.5A	Fastest	1200kHz	16ms	45ms
Default	0h	1h	1h	5h	1h	2h
Combination	2Dh				12h	
Resolution	NA	0.5A	NA	NA	2ms	5ms
0H	PSM	1.0A	Fast	600kHz	2ms	0ms
1H	FCCM	1.5A	Normal	715kHz	4ms	5ms
2H		2.0A	Slow	800kHz	6ms	10ms
3H		2.5A	Slowest	900kHz	8ms	15ms
4H				1000kHz	10ms	20ms
5H				1200kHz	12ms	25ms

	PAVDD Config					
	EN_PAVDD FCCM	PAVDD Current Limit	PAVDD LX FSR	PAVDD LX Frequency	PAVDD Soft-start Time	PAVDD Delay Time
6H					14ms	30ms
7H					16ms	35ms
8H						40ms
9H						45ms

	NAVDD Config					
	NAVDD Soft-start Time	NAVDD Delay Time	EN_NAVD D FCCM	NAVDD Current Limit	NAVDD LXI SR	NAVDD LXI Frequency
Data Address	12h		13h			
Bits	[6:4]	[3:0]	[7]	[6:5]	[4:3]	[2:0]
Min	2ms	0ms	NA	1A	Slow	600kHz
Max	16ms	45ms	NA	2.5A	Fastest	1200kHz
Default	1h	2h	0h	0h	1h	5h
Combination	12h		0Dh			
Resolution	2ms	5ms	NA	0.5A	NA	NA
0H	2ms	0ms	PSM	1.5A	Fast	600kHz
1H	4ms	5ms	FCCM	2.0A	Normal	715kHz
2H	6ms	10ms		2.5A	Slow	800kHz
3H	8ms	15ms		3.0A	Slowest	900kHz
4H	10ms	20ms				1000kHz
5H	12ms	25ms				1200kHz
6H	14ms	30ms				
7H	16ms	35ms				
8H		40ms				
9H		45ms				

	VGH/VGL Config						
	LXH/N SR	SIBO VINP Source	LXP/LXN Frequency	VGH Soft-start Time	VGH Delay Time	VGL Soft-start Time	VGL Delay Time
Data Address	14h			15h		16h	
Bits	[5:4]	[3]	[2:0]	[5:4]	[3:0]	[5:4]	[3:0]
Min	Slow	NA	600kHz	4ms	0ms	4ms	0ms
Max	Fastest	NA	1200kHz	12ms	45ms	12ms	45ms
Default	1h	0h	5h	0h	5h	0h	3h
Combination	15h			05h		03h	
Resolution	NA	NA	NA	NA	5ms	NA	5ms
0H	Fast	VIN	600kHz	4ms	0ms	4ms	0ms
1H	Normal	PAVDD	715kHz	6ms	5ms	6ms	5ms
2H	Slow		800kHz	8ms	10ms	8ms	10ms
3H	Slowest		900kHz	12ms	15ms	12ms	15ms
4H			1000kHz		20ms		20ms
5H			1200kHz		25ms		25ms
6H					30ms		30ms
7H					35ms		35ms

	VGH/VGL Config						
	LXH/N SR	SIBO VINP Source	LXP/LXN Frequency	VGH Soft-start Time	VGH Delay Time	VGL Soft-start Time	VGL Delay Time
8H					40ms		40ms
9H					45ms		45ms
AH					50ms		50ms
BH					55ms		55ms
CH					60ms		60ms
DH					65ms		65ms
EH					70ms		70ms
FH					75ms		75ms

	VCORE Config				VIO Config			
	VCORE LXB1 Slew Rate	VCORE LXB1 Frequency	VCORE Soft-start Time	VCORE Delay Time	VIO LXB2 Slew Rate	VIO LXB2 Frequency	VIO Soft-start Time	VIO Delay Time
Data Address	17h		18h		19h		1Ah	
Bits	[4:3]	[2:0]	[5:4]	[3:0]	[4:3]	[2:0]	[5:4]	[3:0]
Min	Slow	600kHz	2ms	0ms	Slow	600kHz	2ms	0ms
Max	Fastest	1200kHz	8ms	45ms	Fastest	1200kHz	8ms	45ms
Default	1h	5h	1h	1h	1h	5h	1h	2h
Combination	0Dh		11h		0Dh		12h	
Resolution	NA	NA	2ms	5ms	NA	NA	2ms	5ms
0H	Fast	600kHz	2ms	0ms	Fast	600kHz	2ms	0ms
1H	Normal	715kHz	4ms	5ms	Normal	715kHz	4ms	5ms
2H	Slow	800kHz	6ms	10ms	Slow	800kHz	6ms	10ms
3H	Slowest	900kHz	8ms	15ms	Slowest	900kHz	8ms	15ms
4H		1000kHz		20ms		1000kHz		20ms
5H		1200kHz		25ms		1200kHz		25ms
6H				30ms				30ms
7H				35ms				35ms
8H				40ms				40ms
9H				45ms				45ms

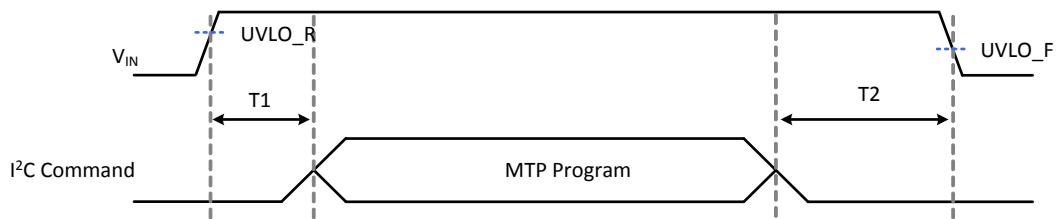
	VCOM Config		RESET Config		Digital VCOM Voltage
	VCOM Pull to GND	VCOM Power off Follow	RESET Power off Control	RESET Delay Time	D-VCOM
Data Address	1Bh				Slave ID=0x9E
Bits	[7]	[6]	[5:4]	[3:0]	[7:1]
Min	NA	NA	NA	0ms	NA
Max	NA	NA	NA	75ms	NA
Default	1h	0h	0h	3h	40h
Combination	83h				40h
Resolution	NA	NA	20ms	5ms	10mV
0H	Disable	Follow UVLO_F	Always High	0ms	VCOM-(10mVx64)
1H	Enable	Follow RESET	20ms	5ms	VCOM-(10mVx63)
2H			40ms	10ms	VCOM-(10mVx62)
3H			60ms	15ms	VCOM-(10mVx61)
4H				20ms	VCOM-(10mVx60)
5H				25ms	VCOM-(10mVx59)
6H				30ms	VCOM-(10mVx58)
7H				35ms	VCOM-(10mVx57)
8H				40ms	VCOM-(10mVx56)
9H				45ms	VCOM-(10mVx55)
AH				50ms	VCOM-(10mVx54)
BH				55ms	VCOM-(10mVx53)
CH				60ms	VCOM-(10mVx52)
DH				65ms	VCOM-(10mVx51)
EH				70ms	VCOM-(10mVx50)
FH				75ms	VCOM-(10mVx49)
10H					VCOM-(10mVx48)
11H					VCOM-(10mVx47)
12H					VCOM-(10mVx46)
13H					VCOM-(10mVx45)
14H					VCOM-(10mVx44)
15H					VCOM-(10mVx43)
16H					VCOM-(10mVx42)
17H					VCOM-(10mVx41)
18H					VCOM-(10mVx40)
19H					VCOM-(10mVx39)
1AH					VCOM-(10mVx38)
1BH					VCOM-(10mVx37)
1CH					VCOM-(10mVx36)
1DH					VCOM-(10mVx35)
1EH					VCOM-(10mVx34)
1FH					VCOM-(10mVx33)
20H					VCOM-(10mVx32)
21H					VCOM-(10mVx31)
22H					VCOM-(10mVx30)
23H					VCOM-(10mVx29)
24H					VCOM-(10mVx28)
25H					VCOM-(10mVx27)
26H					VCOM-(10mVx26)
27H					VCOM-(10mVx25)
28H					VCOM-(10mVx24)
29H					VCOM-(10mVx23)

	VCOM Config		RESET Config		Digital VCOM Voltage
	VCOM Pull to GND	VCOM Power off Follow	RESET Power off Control	RESET Delay Time	D-VCOM
2AH					VCOM-(10mVx22)
2BH					VCOM-(10mVx21)
2CH					VCOM-(10mVx20)
2DH					VCOM-(10mVx19)
2EH					VCOM-(10mVx18)
2FH					VCOM-(10mVx17)
30H					VCOM-(10mVx16)
31H					VCOM-(10mVx15)
32H					VCOM-(10mVx14)
33H					VCOM-(10mVx13)
34H					VCOM-(10mVx12)
35H					VCOM-(10mVx11)
36H					VCOM-(10mVx10)
37H					VCOM-(10mVx9)
38H					VCOM-(10mVx8)
39H					VCOM-(10mVx7)
3AH					VCOM-(10mVx6)
3BH					VCOM-(10mVx5)
3CH					VCOM-(10mVx4)
3DH					VCOM-(10mVx3)
3EH					VCOM-(10mVx2)
3FH					VCOM-(10mVx1)
40H					VCOM
41H					VCOM+(10mVx1)
42H					VCOM+(10mVx2)
43H					VCOM+(10mVx3)
44H					VCOM+(10mVx4)
45H					VCOM+(10mVx5)
46H					VCOM+(10mVx6)
47H					VCOM+(10mVx7)
48H					VCOM+(10mVx8)
49H					VCOM+(10mVx9)
4AH					VCOM+(10mVx10)
4BH					VCOM+(10mVx11)
4CH					VCOM+(10mVx12)
4DH					VCOM+(10mVx13)
4EH					VCOM+(10mVx14)
4FH					VCOM+(10mVx15)
50H					VCOM+(10mVx16)
51H					VCOM+(10mVx17)
52H					VCOM+(10mVx18)
53H					VCOM+(10mVx19)
54H					VCOM+(10mVx20)
55H					VCOM+(10mVx21)
56H					VCOM+(10mVx22)
57H					VCOM+(10mVx23)
58H					VCOM+(10mVx24)
59H					VCOM+(10mVx25)
5AH					VCOM+(10mVx26)
5BH					VCOM+(10mVx27)

	VCOM Config		RESET Config		Digital VCOM Voltage
	VCOM Pull to GND	VCOM Power off Follow	RESET Power off Control	RESET Delay Time	D-VCOM
5CH					VCOM+(10mVx28)
5DH					VCOM+(10mVx29)
5EH					VCOM+(10mVx30)
5FH					VCOM+(10mVx31)
60H					VCOM+(10mVx32)
61H					VCOM+(10mVx33)
62H					VCOM+(10mVx34)
63H					VCOM+(10mVx35)
64H					VCOM+(10mVx36)
65H					VCOM+(10mVx37)
66H					VCOM+(10mVx38)
67H					VCOM+(10mVx39)
68H					VCOM+(10mVx40)
69H					VCOM+(10mVx41)
6AH					VCOM+(10mVx42)
6BH					VCOM+(10mVx43)
6CH					VCOM+(10mVx44)
6DH					VCOM+(10mVx45)
6EH					VCOM+(10mVx46)
6FH					VCOM+(10mVx47)
70H					VCOM+(10mVx48)
71H					VCOM+(10mVx49)
72H					VCOM+(10mVx50)
73H					VCOM+(10mVx51)
74H					VCOM+(10mVx52)
75H					VCOM+(10mVx53)
76H					VCOM+(10mVx54)
77H					VCOM+(10mVx55)
78H					VCOM+(10mVx56)
79H					VCOM+(10mVx57)
7AH					VCOM+(10mVx58)
7BH					VCOM+(10mVx59)
7CH					VCOM+(10mVx60)
7DH					VCOM+(10mVx61)
7EH					VCOM+(10mVx62)
7FH					VCOM+(10mVx63)

MTP Program Sequence for Single Chip

MTP program timing sequence



Write Timing :

00h~1Bh : $T1 = 50\text{ms}$, $T2 = 120\text{ms}$

D-VCOM : $T1 = 50\text{ms}$, $T2 = 15\text{ms}$

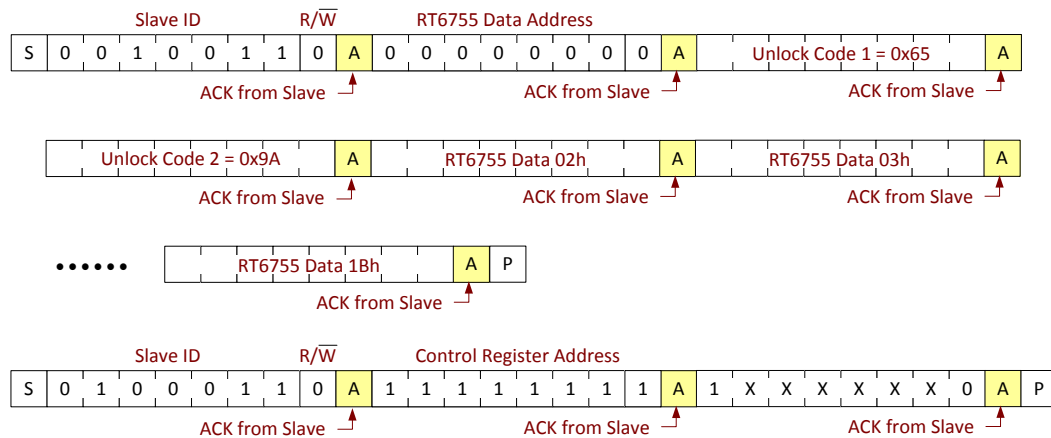
Read Timing:

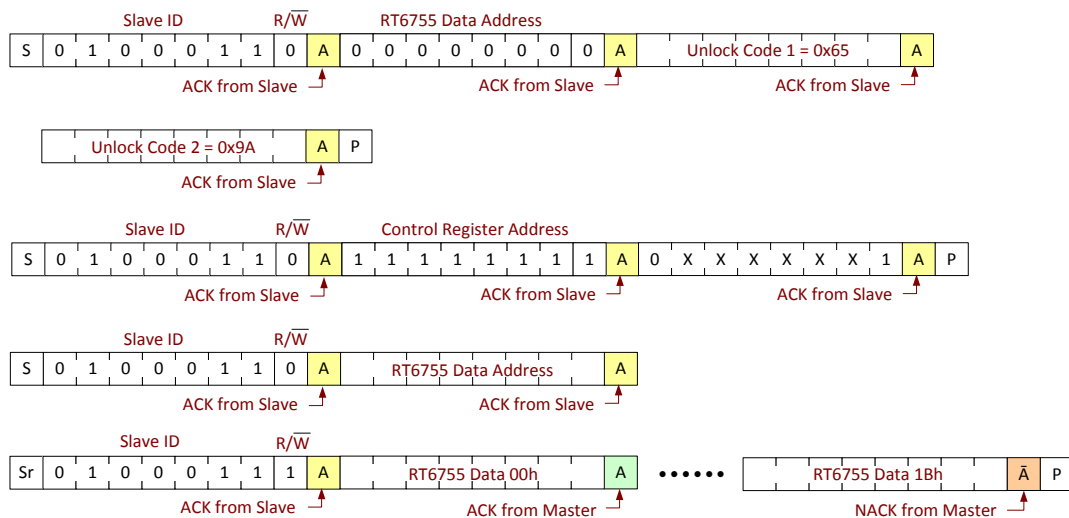
$T1 = 50\text{ms}$, $T2 = 10\text{ms}$

$f_{SCL} = 400\text{kHz}$

I²C Protocol for MTP Program

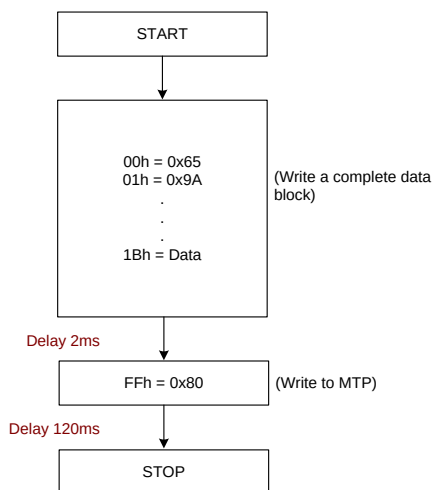
I²C Write Timing Sequence



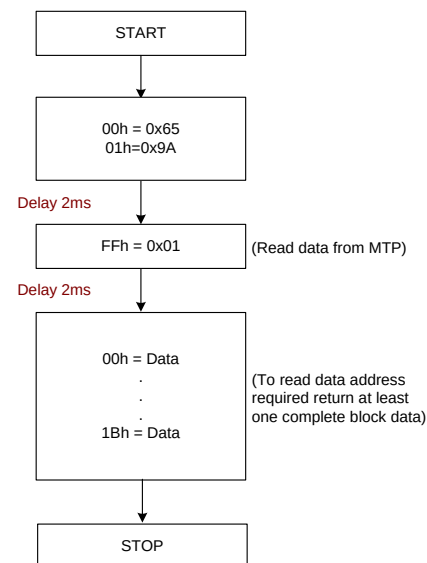
I²C Read Timing Sequence

Read/Write MTP Flow Chat

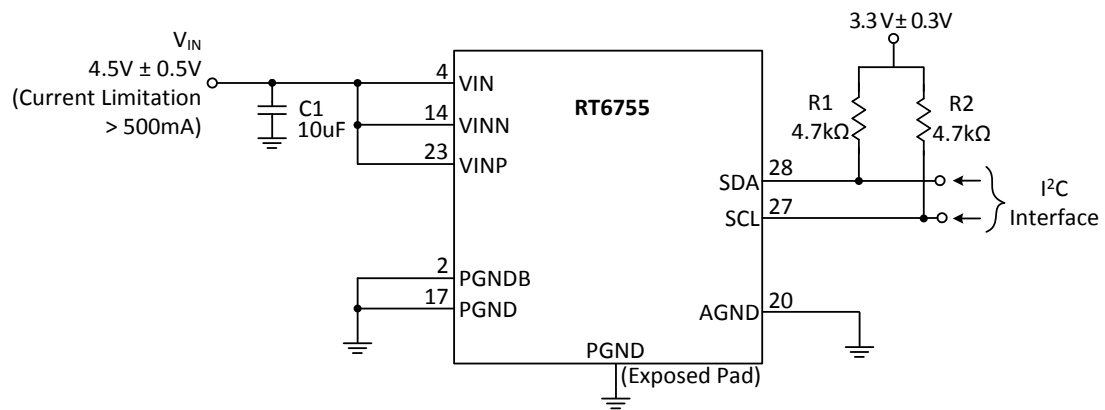
Write Flow



Read Flow



MTP Program Application Circuit for Single Chip



Thermal Considerations

The junction temperature should never exceed the absolute maximum junction temperature $T_{J(MAX)}$, listed under Absolute Maximum Ratings, to avoid permanent damage to the device. The maximum allowable power dissipation depends on the thermal resistance of the IC package, the PCB layout, the rate of surrounding airflow, and the difference between the junction and ambient temperatures. The maximum power dissipation can be calculated using the following formula :

$$P_{D(MAX)} = (T_{J(MAX)} - T_A) / \theta_{JA}$$

where $T_{J(MAX)}$ is the maximum junction temperature, T_A is the ambient temperature, and θ_{JA} is the junction-to-ambient thermal resistance.

For continuous operation, the maximum operating junction temperature indicated under Recommended Operating Conditions is 125°C. The junction-to-ambient thermal resistance, θ_{JA} , is highly package dependent. For a UQFN-28TL 3.5x3.5 package, the thermal resistance, θ_{JA} , is 31.2°C/W on a standard JEDEC 51-7 high effective-thermal-conductivity four-layer test board. The maximum power dissipation at $T_A = 25^\circ\text{C}$ can be calculated as below :

$$P_{D(MAX)} = (125^\circ\text{C} - 25^\circ\text{C}) / (31.2^\circ\text{C/W}) = 3.2\text{W for a UQFN-28TL 3.5x3.5 package.}$$

The maximum power dissipation depends on the operating ambient temperature for the fixed $T_{J(MAX)}$ and the thermal resistance, θ_{JA} . The derating curves in Figure 2 allows the designer to see the effect of rising ambient temperature on the maximum power dissipation.

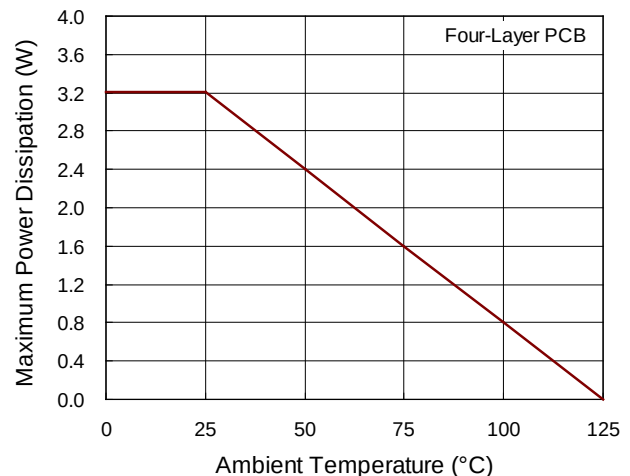


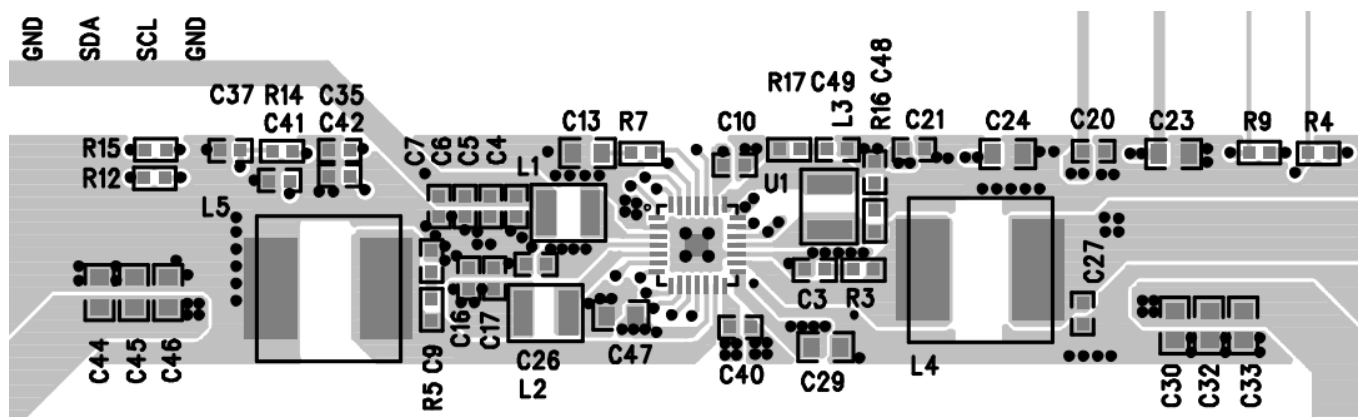
Figure 5. Derating Curve of Maximum Power Dissipation

Layout Consideration

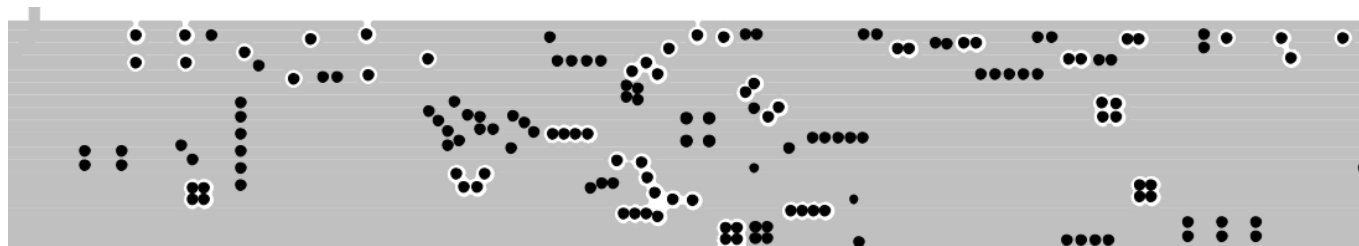
For the best performance of the RT6755. The following descriptions are the guidelines for better PCB layout :

- ▶ The power components such as inductor(L1, L2, L3, L4, L5), input cap(C10, C26, C27, C40) and output cap(C4, C5, C6, C7, C16, C17, C20, C21, C23, C24, C29, C30, C31, C32, C44, C45, C46, C47) must be placed as close as possible to reduce power loop. The PCB trace between power components must be as short and wide as possible.
- ▶ Minimize the size of the LX, LXI, LXB1, LXB2, LXP, LXN node and keep it wide and short. Keep the LX, LXI, LXB1, LXB2, LXP and LXN node away from those sensing pins (COMP, VCOM, NEG) and analog ground.
- ▶ The power ground (PGNDB, PGND) consists of input and output capacitor grounds.
- ▶ The compensation circuit (R7, C13) should be kept away from the power loops and should be shielded with a ground trace to prevent any noise coupling. Place the compensation components as close as possible to COMP pin.
- ▶ The exposed pad of the chip should be connected to a ground plane for thermal consideration.

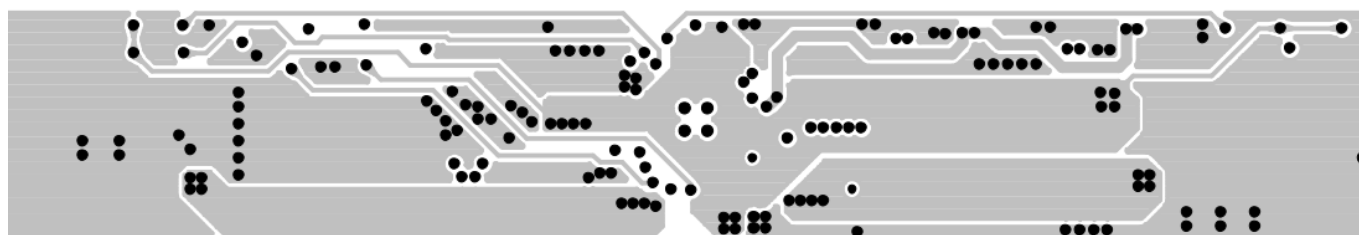
Top Layer:



Second Layer:



Third Layer:



Bottom Layer:

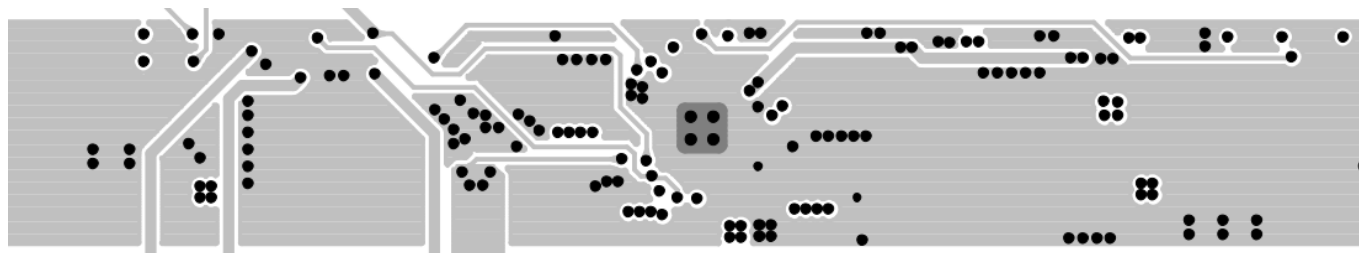
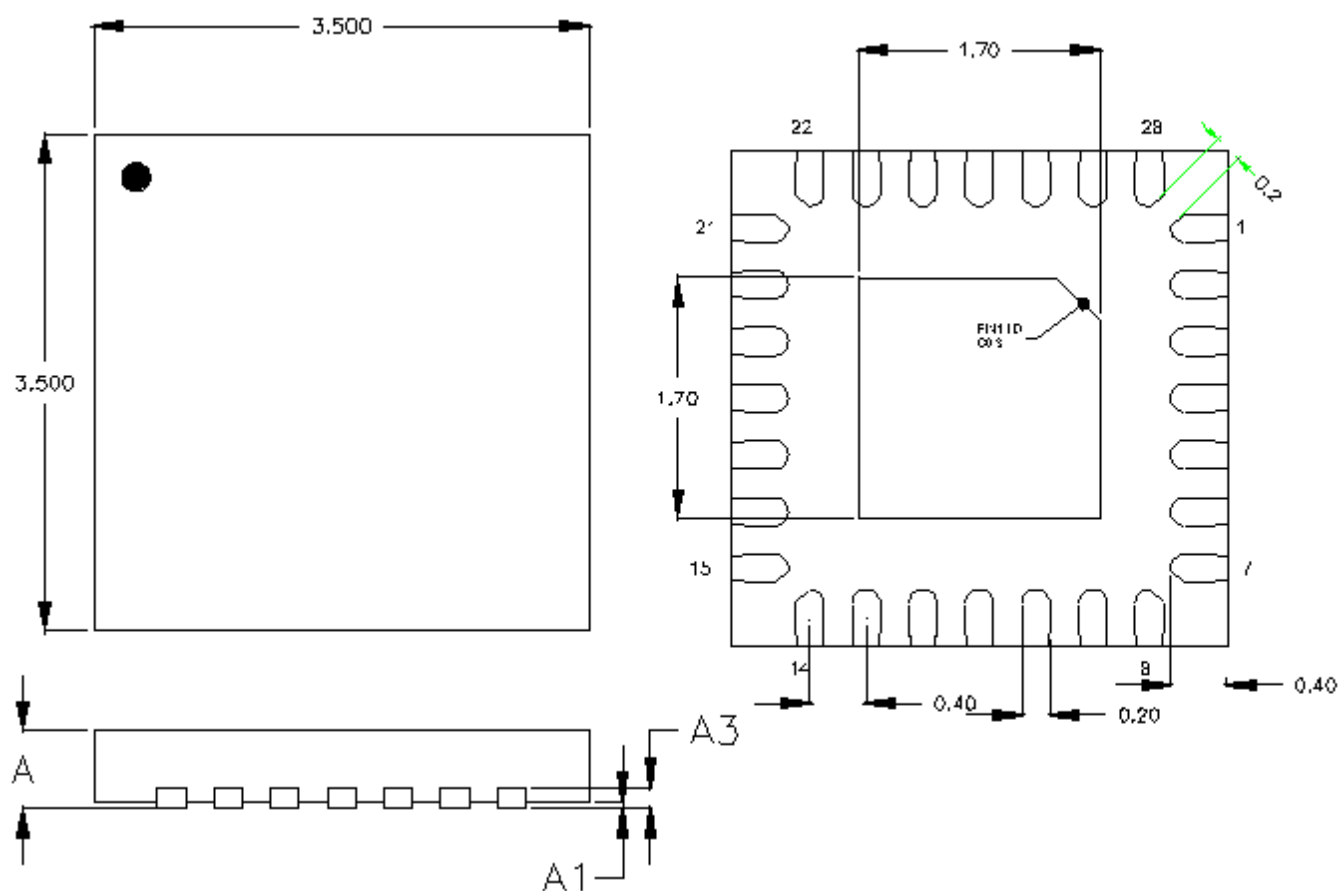


Figure 6. PCB Layout Guide

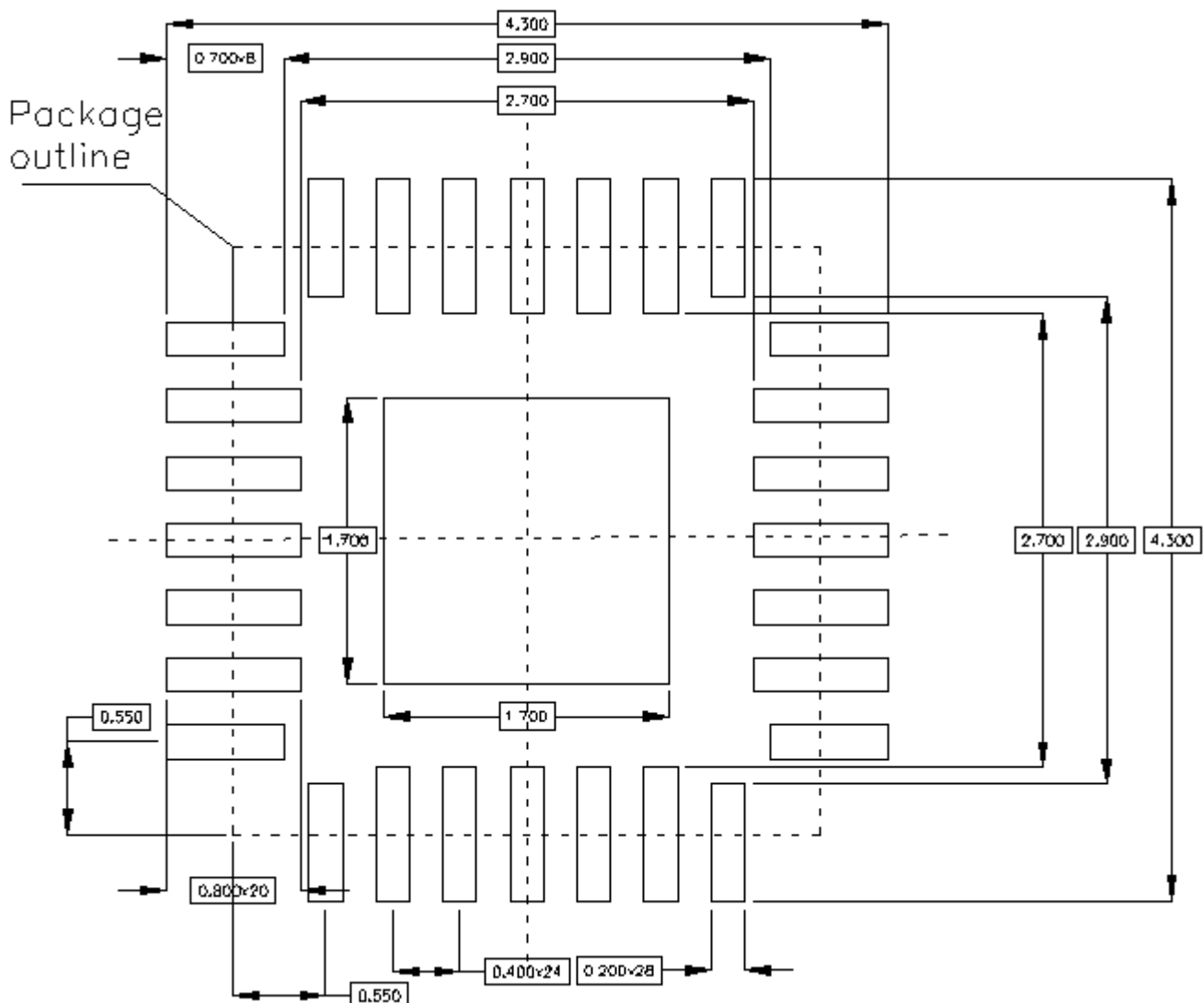
Outline Dimension



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	0.500	0.600	0.020	0.024
A1	0.000	0.050	0.000	0.002
A3	0.100	0.200	0.004	0.008

U-Type 28TL QFN 3.5x3.5 (FC) Package

Footprint Information



Package	Number of Pin	Tolerance
VW/U/XQFN3.5x3.5-28T(FC)	28	±0.05

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DS6755-P03 December 2021

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Datasheet Revision History

Version	Date	Item	Description
P00	2021/2/22		First Edition
P01	2021/7/26	General Description Ordering Information Pin Configuration Typical Application Circuit Absolute Maximum Ratings Electrical Characteristics Application Information Outline Dimension Footprint Information	Modify
P02	2021/9/24	Electrical Characteristics	Modify
P03	2021/12/29	Typical Application Circuit Timing Diagram Typical Operating Characteristics Applications Information	Modify